Consciousness Trilogy

Inseparability, Gravity, and the Open Explanatory Gap

Complete Collected Volume

Papers I, II, and III

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Abstract

This volume collects the three papers of the *Consciousness Trilogy*, an algebraic and information-geometric programme that recasts the hard problem of consciousness as a question about the independent deformability of geometric and first-person structural data.

The trilogy develops a three-step logical chain:

1. **Paper I: Inseparability.** Within a Type III_1 algebraic setting, emergent geometry and modular flow are shown not to be independently deformable. The result is a structural inseparability theorem, not a reductive identity claim, and the explanatory gap is relocated from an alleged production failure to the problem of self-description by finite observers.

2. **Paper II: Higher-Dimensional Consistency.** In the CHM small-ball framework, the higher-dimensional modular datum is identified as the full family of ball modular Hamiltonians across centers, radii, and timelike directions. Compatibility with one geometry yields a conditional leading-order consistency relation in the CHM small-ball regime; it does not by itself prove a general gravitational uniqueness claim.

3. **Paper III: Open Explanatory Gap.** Finite self-description is studied through conditional arithmetic self-models. The former local minimax drift-floor theorem is withdrawn, and the residual explanatory gap remains open rather than quantitatively closed by a nonzero floor.

Taken together, the three papers propose a coherent programme in which the hard problem's standard independent-describability presupposition is not available in the relevant algebraic setting, higher-dimensional consistency can be analyzed in a CHM local regime, and the remaining finite self-description gap is explicitly left open.

Keywords: algebraic quantum field theory, Tomita–Takesaki modular theory, Guido–Longo–Wiesbrock reconstruction, CHM ball Hamiltonians, linearized gravity, Fisher information geometry, self-description, explanatory gap

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The Inseparability of Geometry and Modular Flow: A Structural Inseparability Theorem and Its Implications for the Hard Problem

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Abstract

The hard problem of consciousness presupposes that physical structure and experiential structure can be specified independently. We show that this presupposition is not available in a precise algebraic setting.

Working with Type III_1 von Neumann algebras, we identify emergent geometry as the relevant mathematical proxy for physical structure and Tomita-Takesaki modular flow as the relevant proxy for first-person temporal structure. The resulting question is whether these two sectors can be independently deformed within an admissible algebraic class. We show first that, working within the admissible class selected by $(U3)$, from the proof-relevant axiom package

$$(U1) + (C) + (Amb) + (CN_{\exists}),$$

one obtains a standard half-sided modular inclusion and hence, via the Guido-Longo-Wiesbrock construction, a canonical conformal skeleton. This yields a conformal-level Weak Inseparability theorem. We then show that the correct non-gauge local modulus is the double-interval cross-ratio family, which determines a canonical modular scale datum

$$\{K_b\}_{b\in(0,1)}.$$

Under the additional conditions $(LImp^1)$, $(CR\text{-}nd)$, and (SFa), this datum induces a Type III-compatible QFIM scale shadow

$$h_b = \mathbf{Q}_{\omega,b}(\partial_b K_b, \partial_b K_b)\, db^2,$$

At the abstract level, this yields a full geometric datum consisting of the conformal skeleton together with the scale family. In the worked examples, where the cross-ratio parameter naturally parametrizes spatial position, this datum determines the Lorentzian metric. In this setting, the full geometric datum and the extended modular data are bidirectionally inseparable.

We verify the additional Phase 2 conditions in two structurally distinct models, the Rindler/free-field chiral half-line model and the Virasoro net. Both yield the universal profile

$$h_b = C(1-b)^{-2}db^2,$$

with model dependence entering only through the overall constant. The philosophical consequence is not that the hard problem has been solved in the ordinary production sense, but that its standard formulation rests on a presupposition—independent describability—that is not available in the relevant algebraic setting. The remaining explanatory gap is thereby relocated from a gap between independently specified domains to a structural limit on self-description by observers embedded in $(\mathcal{A}, \omega)$.

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1 The Hidden Presupposition of the Hard Problem

The hard problem of consciousness is usually presented as a question about production: how do physical processes give rise to subjective experience? In that form, the problem appears deeper than the familiar "easy problems" because those concern discrimination, report, integration, control, and other functional capacities, whereas the hard problem concerns the existence of a first-person point of view and the felt character of experience itself. This contrast, introduced most influentially by Chalmers [4], has shaped almost every major debate in contemporary philosophy of mind.

Yet the canonical formulation contains a strong presupposition that is rarely isolated and almost never challenged at the level of formal structure. The presupposition is that physical structure and experiential structure can be specified independently. One first gives a complete account of the physical facts, in third-person terms, and then asks why those facts are accompanied by experience, or by this experience rather than another. Without this prior separability, the production question cannot even be posed in its usual form. The hard problem therefore depends not only on a gap between two domains, but on the prior availability of those domains as independently describable.

This presupposition is shared by positions that otherwise disagree sharply. Functionalism and mainstream physicalism accept it and then attempt to dissolve the residual gap by appeal to organization, role, or supervenience. Panpsychism accepts it as well, but seeks to avoid brute emergence by distributing proto-experiential or experiential properties more fundamentally. Illusionism also inherits the same starting point: it grants the independent specification of physical structure and then argues that what seemed left over on the experiential side is a cognitive misdescription. Integrated Information Theory takes a different route, but still treats consciousness as something to be attached to or read off from an already specified structure by means of an additional quantity. In all of these cases, the debate concerns how the bridge should be built. The deeper assumption that there are two independently specifiable shores is left intact.

The present paper takes aim at that assumption. It does not begin by proposing a new psychophysical law, a new emergent mechanism, or a new primitive consciousness quantity. It begins by asking whether the independent-describability presupposition survives translation into a precise mathematical framework. If "physical structure" is represented not by an unconstrained intuitive picture of matter but by an emergent geometric structure, and if "first-person time" is represented by modular flow in the sense of Tomita-Takesaki theory, then the hard problem can be reformulated as a structural question: can these two sectors be independently varied inside a single admissible algebraic setting?

The central claim of this paper is that, under a precise Type III_1 algebraic framework, they cannot. At the conformal level, emergent causal geometry and conformal modular data are already mutually rigid. With three additional Phase 2 inputs, the same becomes true at the full Lorentzian level: complete geometry and extended modular data cannot be independently deformed. This does not amount to the claim that the hard problem has simply been "solved." It amounts to a more exact and, in one sense, more radical thesis: the standard production form of the hard problem depends on a presupposition that is not available in the relevant mathematical setting.

This is why the main theorem of the paper is an inseparability theorem rather than a reduction theorem. The point is not that one side reduces to the other, nor that one side causally generates the other, but that both emerge as two readings of one underlying algebraic organization. In philosophical terms, the paper argues that the usual mind-body contrast is not fun-

damental enough to underwrite the standard hard-problem question. In mathematical terms, it shows that the admissible deformations of geometry and modular structure do not factor into two independent sectors.

The argument unfolds in three stages. First, we translate the relevant philosophical notions into precise mathematical objects: emergent geometry for the object-side physical structure, and modular flow for first-person temporal structure. Second, we prove the conditional Phase 1 result: working with the carrier specified by $(U3)$, from the proof-relevant axiom package

$$(U1) + (C) + (Amb) + (CN_{\exists})$$

one canonically obtains a conformal skeleton, together with a conformal-level inseparability theorem. Third, we explain how the full metric scale is added by the cross-ratio modular datum and its Type III QFIM shadow, yielding a conditional full inseparability theorem under the additional Phase 2 package

$$(\text{LImp}^1) + (\text{CR-nd}) + (\text{SFa}).$$

Two worked examples, one based on the Rindler/free-field chiral half-line model and one on the Virasoro net, show that the Phase 2 conditions are not merely formal placeholders but are realized in structurally distinct settings.

The resulting philosophical payoff is twofold. First, the hard problem is reframed: the primary error lies not in failing to build a bridge, but in assuming that the bridge must connect independently specifiable domains. Second, the residual explanatory gap is relocated rather than denied. Once independent describability fails, the remaining question is no longer how matter produces experience, but why a finite observer embedded in a single algebraic structure cannot completely internalize the relation between structural invariants and lived articulation. That second issue belongs to the limits of self-description, not to a production gap between two ontologically detached realms.

The rest of the paper is organized as follows. Section 2 identifies the mathematical counterparts of physical structure and first-person time, and formulates the independent-describability issue as a question of independent deformability. Section 3 states the algebraic framework and axioms. Section 4 proves the conditional Phase 1 result: from Paper I's shell interface and dynamical package to a canonical conformal skeleton and Weak Inseparability. Section 5 develops the Phase 2 completion via the cross-ratio modular scale datum and the full inseparability theorems. Section 6 identifies abstract sufficient conditions for the chart axiom $(LImp^1)$: for Diff-covariant nets it is fully discharged, and for non-Diff-covariant nets the remaining burden is isolated to transportability and regularity assumptions. Section 7 verifies the remaining Phase 2 conditions in two worked examples. Section 8 then returns to the hard problem directly and argues that the theorem shows its standard formulation rests on a presupposition not available in the stated algebraic setting, while relocating the remaining explanatory gap to the self-referential horizon of finite observers. Section 9 discusses the broader mathematical and physical significance, and Section 10 states the explicit non-claims and open problems.

1.1 Logical Roadmap

The argument has two phases, each with a clear input-output structure.

Phase 1 (Sections 3–4). Input: the axiom package $(U1) + (C) + (Amb) + (CN_{\exists})$, working with the carrier specified by $(U3)$. Output: a canonical conformal skeleton via the Guido-Longo-Wiesbrock reconstruction, and a conformal-level Weak Inseparability theorem (Theorem 4.8). The key new step is that half-sided modular inclusion is *derived* from the modular-decoherence scaling axiom (C), not assumed.

Phase 2 (Section 5). Input: the additional conditions $(LImp^1)+(CR\text{-}nd)+(SFa)$. Output: a full geometric datum (conformal skeleton + scale shadow) and bidirectional Full Inseparability (Theorem 5.7). The key new object is the cross-ratio modular scale datum $\{K_b\}$ and its Araki-Hessian QFIM shadow h_b.

Section 6 provides abstract sufficient conditions for $(LImp^1)$. Section 7 verifies all Phase 2 conditions in two worked examples. Sections 8–10 draw philosophical consequences, discuss broader significance, and state open problems.

2 Mathematical Identification: Geometry as Physics, Modular Flow as First-Person Time

The philosophical argument of this paper rests on a mathematical theorem. For the theorem to bear on the hard problem, the translation between philosophical concepts and mathematical objects must be made explicit and defended. This section fixes the three identifications that underwrite the rest of the paper: emergent geometry as the mathematical content of physical structure, Tomita-Takesaki modular flow as the mathematical content of first-person time, and independent deformability as the mathematical translation of independent describability.

2.1 Emergent Geometry as Physical Structure

In the standard formulation of the hard problem, "physical structure" is left deliberately broad. It may refer to microphysical states, functional organization, or spacetime geometry, depending on the author. For the present purposes, the relevant notion of physical structure is not an arbitrary state description but the emergent spacetime geometry associated with a quantum-algebraic datum.

The reason for this choice is not merely technical convenience. In the algebraic approach to quantum field theory, spacetime is not a fixed background container. It is recovered from the algebraic relations among observables. The programme initiated by Wiesbrock and developed by Guido, Longo, and others has shown that, under appropriate conditions, the inclusion structure of von Neumann algebras determines a conformal geometry [17, 5], and in favorable cases a full Lorentzian metric. In this setting, "physical structure" is not something given prior to the algebra. It is something read off from the algebra.

This is the correct level of description for the hard problem, because the hard problem is not about the relation between consciousness and some particular physical substrate such as neurons or silicon. It is about the relation between the subjective and the objective as structural categories. If the objective-physical side is itself an emergent reading of algebraic data, then the question of whether it can be independently specified becomes a question about the algebraic structure, not about a particular material realization.

We therefore identify:

$$\text{physical structure} \longleftrightarrow \text{Geom}(\mathcal{N}, \omega),$$

where Geom denotes the emergent geometry extracted from a family $\mathcal{N}$ of von Neumann algebras in a state ω. At the Phase 1 level, this is a conformal skeleton. At the full level, it includes the metric scale.

2.2 Modular Flow as First-Person Time

The second identification is less familiar in the philosophy of mind but is equally well motivated from the algebraic side.

Given a von Neumann algebra $\mathcal{A}$ and a faithful normal state ω, the Tomita-Takesaki theorem [14] produces a canonical one-parameter automorphism group σ_t^ω of $\mathcal{A}$, the modular automorphism group. This group has three properties that make it the natural mathematical counterpart of first-person temporal structure.

First, uniqueness. The modular flow is the unique automorphism group satisfying the KMS condition at inverse temperature $\beta = 1$ with respect to ω. No external clock or background time parameter is needed to define it. It is intrinsic to the pair $(\mathcal{A}, \omega)$.

Second, algebraic relativity. Different choices of algebra $\mathcal{A}$ within the same ambient structure yield different modular flows. This is not a defect. It is the mathematical expression of the fact that different observational perspectives give rise to different temporal experiences. The Unruh effect is the most familiar physical instance [15]: an accelerated observer's modular flow differs from that of an inertial observer, and the difference is experienced as a thermal bath.

Third, structural inseparability at the operator level. The Tomita-Takesaki polar decomposition

$$S_\Omega = J_\Omega \Delta_\Omega^{1/2}$$

is unique. The modular operator Δ and the modular conjugation J are not independently choosable components. One cannot have Δ without J, nor J without Δ. Since Δ^{it} generates the modular flow and J implements the exchange between the algebra and its commutant, this already hints at the inseparability that the main theorem will establish at the geometric level: the temporal structure (Δ) and the spatial/duality structure (J) are two factors of a single operator.

We therefore identify:

$$\text{first-person time} \longleftrightarrow \sigma_t^\omega = \Delta_\omega^{it}(\cdot)\Delta_\omega^{-it}.$$

More precisely, the full modular sector includes not only the one-parameter flow but also the modular Hamiltonian $K_\omega = -\ln \Delta_\omega$, the modular conjugation J_ω, and, at the extended level, the local modular scale data $\{K_b\}$ introduced in Phase 2.

2.3 From Independent Describability to Independent Deformability

With the two identifications in place, the philosophical question can be stated mathematically.

The hard problem, in its standard form, presupposes that one can give a complete specification of the physical facts and then ask, as a further question, why those facts are accompanied by experience. Translated into the present framework, this becomes: can one specify $\mathrm{Geom}(\mathcal{N}, \omega)$ without thereby fixing $\mathrm{Mod}(\mathcal{N}, \omega)$, or conversely?

The natural mathematical formalization of "independent specification" is independent deformability. Two structures are independently describable, in the relevant sense, if and only if there exist admissible deformations of the underlying algebraic data that change one while leaving the other invariant. If no such deformation exists, then specifying one structure automatically constrains the other, and the two cannot be treated as independently given.

This is a precise, falsifiable mathematical question. It does not depend on intuitions about zombies, inverted qualia, or conceivability arguments. It depends on the rigidity properties of the map

$$(\mathcal{N}, \omega) \longmapsto \big(\mathrm{Geom}(\mathcal{N}, \omega),\ \mathrm{Mod}(\mathcal{N}, \omega)\big)$$

within the admissible class defined by the axiom package.
The hard problem, so translated, becomes:

> Does there exist an admissible deformation of $(\mathcal{N}, \omega)$ that changes Geom while preserving Mod, or vice versa?

The inseparability theorems of this paper answer: no. At the conformal level (Phase 1), the answer is unconditional within the axiom class. At the full scale-sensitive level (Phase 2), the answer is conditional on three regularity inputs that are verified in two worked examples.

2.4 What This Translation Does and Does Not Claim

Two caveats are essential.

First, the identification of modular flow with first-person time is a structural correspondence, not an identity claim. We do not assert that modular flow "is" consciousness in any reductive sense. We assert that modular flow is the correct mathematical object that plays the structural role assigned to first-person temporal experience in the hard-problem debate. The correspondence is justified by the three properties listed above: intrinsicness, perspectival relativity, and operator-level inseparability from spatial structure.

Second, the identification of emergent geometry with physical structure is restricted to the level of spacetime geometry. It does not claim to capture all of physics. In particular, it does not address the relation between consciousness and specific material substrates such as neural tissue. The claim is that at the level of structural categories relevant to the hard problem, emergent geometry is the right proxy for the objective-physical side. Whether this extends to more fine-grained physical descriptions is an open question that lies beyond the scope of the present paper.

3 Algebraic Framework and Axioms

We now state the minimal algebraic framework in which the failure of independent deformability can be proved. The point of this section is not to give a general introduction to Type III_1 factors or to algebraic quantum field theory. It is to isolate the exact package of assumptions needed for the Phase 1 theorem and to indicate, in each case, why that assumption belongs in the present hard-problem-led setting.

3.1 Ambient Datum and Admissible Object

We work in the standard form of an ambient von Neumann algebra

$$(\mathcal{H}_U, \mathcal{A}_U, \Omega),$$

where Ω is cyclic and separating for $\mathcal{A}_U$. This is the minimal regularity needed for Tomita-Takesaki theory and is satisfied in all standard AQFT models [6]. Inside $\mathcal{A}_U$ we select an accessible von Neumann subalgebra

$$\mathcal{A}_c \subset \mathcal{A}_U,$$

and define the reference state

$$\omega(X) := \langle \Omega, X\Omega \rangle.$$

The pair

$$(\mathcal{A}_c, \omega)$$

is the basic object of the theory. In the intended applications, $\mathcal{A}_c$ is of Type III_1. This is not an arbitrary technical preference. Type III_1 is the natural habitat of local quantum field theoretic algebras [6], and it is precisely the setting in which modular theory is rich enough to support the present route from algebraic accessibility data to emergent geometry.

The ambient algebra $\mathcal{A}_U$ should be thought of as the unreduced field of possible observables, while $\mathcal{A}_c$ is the algebra actually stabilized or selected by the accessibility criterion. The question of the paper is then asked not about the entire ambient structure at once, but about what can be extracted from the selected pair $(\mathcal{A}_c, \omega)$.

3.2 Inherited Accessibility Conditions

The first two carrier assumptions are motivated by the HAFF setting [8], but are stated here as independent Paper I inputs.

(U1) Faithfulness. The restriction of the reference state to the accessible algebra is faithful:

$$\omega|_{\mathcal{A}_c} \text{ is faithful.}$$

Equivalently, Ω is cyclic and separating for $\mathcal{A}_c$.

This is the minimal condition that allows modular theory to be applied to $(\mathcal{A}_c, \omega)$. In the present philosophical translation, $(U1)$ says that the allegedly "first-person" sector is not attached to a degenerate algebraic viewpoint. The accessible algebra must support a genuine intrinsic modular structure.

(U3) Stable carrier. The accessible algebra is supplied as a von Neumann subalgebra that is faithful and forward-stable under the decoherence semigroup:

$$\mathcal{E}_t(\mathcal{A}_c) \subset \mathcal{A}_c, \qquad \forall t \geq 0,$$

Here $\{\mathcal{E}_t\}_{t\geq 0}$ is the decoherence semigroup acting on the ambient algebra: a strongly continuous one-parameter semigroup of normal completely positive unital maps on $\mathcal{A}_U$ satisfying the Markov property $\mathcal{E}_0 = \mathrm{id}$ and $\mathcal{E}_{s+t} = \mathcal{E}_s \circ \mathcal{E}_t$. Its physical role is to model the progressive loss of coherence between the accessible algebra and its environment. No identity $\mathcal{A}_c = \mathrm{Fix}(\mathcal{E})$, and no uniqueness or maximality property, is assumed here. (For the motivating construction, see [8].)

The role of $(U3)$ is only to specify the carrier on which the modular construction is performed. The Phase 1 proofs use forward stability, not a selection or uniqueness theorem. Canonical selection of $\mathcal{A}_c$ from a larger registry of accessible effects, probe channels, records, or exact fixed points is a separate interface problem.

Throughout this paper, we work with a carrier satisfying $(U3)$. No claim below promotes that supplied carrier to the unique accessible algebra.

3.3 The New Dynamical Axiom

The crucial new input is the scaling relation between decoherence and modular evolution.

(C) Modular-decoherence scaling covariance. For all $s \in \mathbb{R}$ and $t \geq 0$,

$$\mathcal{E}_t \circ \mathrm{Ad}\,\Delta_{\mathcal{A}_c}^{is} = \mathrm{Ad}\,\Delta_{\mathcal{A}_c}^{is} \circ \mathcal{E}_{e^{2\pi s}t}.$$

Here $\mathrm{Ad}\,\Delta_{\mathcal{A}_c}^{is}$ denotes the outer conjugation action of the modular unitary on the ambient operator algebra.

This is the only genuinely new dynamical axiom introduced in Phase 1. Its content is that modular evolution does not merely preserve the accessible algebra tautologically. It rescales the effective decoherence depth. In the language of the hard problem, (C) is the first place where the object-side and act-side structures are explicitly tied together. It is therefore the real replacement for the earlier non-scrambling intuition.

Remark 3.1 (Physical plausibility of (C)). Axiom (C) holds whenever the modular flow of the accessible algebra acts as a dilation on the decoherence time parameter. This is the case in the Bisognano-Wichmann setting, where modular flow is a Lorentz boost and decoherence depth scales with the boost parameter, and more generally in any Rindler-type wedge algebra. The axiom is a nontrivial dynamical input: it ties the intrinsic modular time of the accessible algebra to the extrinsic decoherence structure of the ambient algebra. Its domain of validity beyond the HAFF setting [8] is an open question.

Remark 3.2 (Finite-dimensional bounded-generator obstruction). There is no nontrivial finite-dimensional realization of (C) with a bounded generator. If $\mathcal{E}_t = e^{t\mathcal{L}}$, differentiating $\mathcal{E}_t\sigma_s = \sigma_s\mathcal{E}_{e^{2\pi s}t}$ at $t = 0$ gives

$$\sigma_s^{-1}\mathcal{L}\sigma_s = e^{2\pi s}\mathcal{L}.$$

Because the modular action is isometric, similarity by σ_s preserves the operator norm. Hence $\|\mathcal{L}\| = e^{2\pi s}\|\mathcal{L}\|$ for every s, forcing $\mathcal{L} = 0$. Thus a nontrivial model must leave the finite-dimensional bounded-generator regime; this remark is an obstruction, not an existence claim for the Type III case.

3.4 Ambient Shell Realization

To turn (C) into a geometric statement, one needs an external family on which the rescaling can act.

(Amb) Operational shell interface. Fix $\varepsilon > 0$. For O in the norm unit ball $\mathrm{Ball}(\mathcal{A}_U)$, define

$$d_\tau(O) := \sup_{0\le t\le\tau} \|\mathcal{E}_t(O) - O\|, \qquad \mathcal{S}_\varepsilon(\tau) := \{O \in \mathrm{Ball}(\mathcal{A}_U) : d_\tau(O) \le \varepsilon\},$$

and supply the ambient shell family by

$$\mathcal{A}(\tau) := W^*\big(\mathcal{A}_c \cup \mathcal{S}_\varepsilon(\tau)\big), \qquad \tau > 0.$$

The ambient modular action $\sigma_s = \mathrm{Ad}\,\Delta_{\mathcal{A}_c}^{is}$ is assumed to act on $\mathcal{A}_U$ by isometric $*$-automorphisms. The following closure and standardness properties are interface inputs:

1. for every $\tau > 0$,
$$\mathcal{A}_c \subset \mathcal{A}(\tau) \subset \mathcal{A}_U;$$
2. if $\tau_1 < \tau_2$, then
$$\mathcal{A}(\tau_1) \supset \mathcal{A}(\tau_2);$$
this nesting also follows directly from $d_{\tau_1} \le d_{\tau_2}$;
3. the accessible algebra is the core limit:
$$\mathcal{A}_c = \bigcap_{\tau>0} \mathcal{A}(\tau);$$
4. Ω is cyclic and separating for every $\mathcal{A}(\tau)$.

The point of (Amb) is to make shell membership and von Neumann closure explicit. It is a Paper I interface input: Foundation I does not derive this membership rule, its core identity, or its standardness. Crucially, modular covariance is not included in (Amb); shell scaling will be derived from the independent dynamical axiom (C).

3.5 Existential Conormality

The final input required for the Phase 1 conformal reconstruction is a second-direction existence condition.

(CN$_\exists$) Existential conormality. There exists some $\mu_0 > 0$ such that, for

$$\mathcal{C}_{\mu_0} := \mathcal{B}'_{\mu_0} \cap \mathcal{A}_c,$$

the vector Ω is cyclic.

The algebras $\mathcal{B}_\mu$ will be constructed from (C) and (Amb) in Section 4; the present point is only that one cyclicity assumption at one scale is enough to close the second direction and trigger the GLW reconstruction.

Philosophically, $(CN_\exists)$ is the minimal extra input needed to pass from a single half-sided modular direction to a full conformal skeleton. It should therefore be read neither as an arbitrary technical patch nor as a hidden second dynamics. It is an existence-type condition ensuring that the accessible algebra contains enough internal relational richness to support the second lightlike direction.

3.6 What Is Explicitly Not Assumed

One older condition is intentionally removed.

(U2) is deleted. The statement that the modular group preserves the accessible algebra,

$$\sigma_t^{\mathcal{A}_c}(\mathcal{A}_c) = \mathcal{A}_c,$$

is automatic from Tomita-Takesaki theory [14] and therefore carries no genuine selection content.

This deletion matters conceptually. The paper does not rest on a tautological "modular stability" assumption. The real content enters through (C), where modular evolution acts outwardly on the shell family rather than merely inwardly on $\mathcal{A}_c$ itself.

It is also worth noting that the stronger exhaustion condition

$$\text{OE}: \quad W^*\Big(\bigcup_{\tau>0} \mathcal{A}(\tau)\Big) = \mathcal{B}(\mathcal{H}_U)$$

is not part of the minimal Phase 1 package. It is only relevant for stronger scalar-core statements that are not needed for the present conformal-level theorem.

3.7 The Minimal Phase 1 Package

We may now summarize the standing assumptions. Throughout the paper we work within the admissible class selected by $(U3)$. The proof-relevant Phase 1 package is:

$$(U1) + (C) + (Amb) + (CN_\exists).$$

The logic of this package is straightforward.

- $(U1)$ makes modular theory available on the accessible algebra.
- (C) ties modular evolution to decoherence depth.
- (Amb) gives that depth a concrete shell realization inside the ambient algebra.
- $(CN_{\exists})$ provides the minimal extra cyclicity needed to reconstruct the second direction and hence the conformal skeleton.

The remainder of Phase 1 consists in proving that this proof-relevant package is already enough to derive a standard half-sided modular inclusion, a canonical GLW Möbius net, and therefore a conformal-level inseparability theorem. That is the task of the next section.

4 From Axioms to Conformal Skeleton (Phase 1)

4.1 From Axiom C to Standard Half-Sided Modular Inclusion

The first structural point of the paper is that the half-sided modular inclusion needed by the Wiesbrock-GLW machinery is not postulated. It is derived conditionally from Paper I's dynamical axiom (C) and shell interface (Amb). In particular, the present route differs sharply from the earlier formulation in which HSMI appeared as an external ansatz. Here it is the output of the Phase 1 assumptions.

Throughout this section we assume $(U1)$, (C), and the explicit shell interface (Amb), and we work with the carrier specified by $(U3)$. We write

$$\sigma_t := \sigma_t^{\mathcal{A}_c}$$

for the modular automorphism group of $(\mathcal{A}_c, \Omega)$ and

$$J := J_{\mathcal{A}_c}$$

for the associated modular conjugation. By (Amb), we are given a decreasing family of ambient shell algebras

$$\mathcal{A}_c \subset \mathcal{A}(\tau) \subset \mathcal{A}_U, \qquad \tau > 0,$$

with

$$\mathcal{A}_c = \bigcap_{\tau>0} \mathcal{A}(\tau),$$

and whose parameter τ is interpreted as a decoherence window.

4.2 Outer Scaling from Modular-Decoherence Covariance

The real content of axiom (C) is that the modular flow of the accessible algebra does not merely preserve $\mathcal{A}_c$ tautologically. Rather, it rescales the ambient shell parameter. This is the first point at which the older $(U2)$-style formulation must be replaced.

Proposition 4.1 (Outer scaling law)**.** *Conditional on the shell interface* (Amb) *and the independent dynamical covariance axiom* (C)*,*

$$\sigma_s(\mathcal{A}(\tau)) = \mathcal{A}(e^{-2\pi s}\tau), \qquad \forall s \in \mathbb{R},\ \tau > 0.$$

Proof. For $O \in \mathrm{Ball}(\mathcal{A}_U)$, axiom (C) and isometry of σ_s give

$$\begin{aligned} d_{e^{-2\pi s}\tau}(\sigma_s O) &= \sup_{0\le t\le e^{-2\pi s}\tau} \|\mathcal{E}_t(\sigma_s O) - \sigma_s O\| \\ &= \sup_{0\le t\le e^{-2\pi s}\tau} \|\mathcal{E}_{e^{2\pi s}t}(O) - O\| \\ &= d_\tau(O), \end{aligned}$$

where the last line uses $u = e^{2\pi s}t$. Thus $\sigma_s \mathcal{S}_\varepsilon(\tau) = \mathcal{S}_\varepsilon(e^{-2\pi s}\tau)$. Since modular flow preserves $\mathcal{A}_c$, applying σ_s to the generating sets and passing to their von Neumann closures proves the equality. □

This result is the correct algebraic replacement for the old non-scrambling intuition: modular time rescales decoherence depth. It is already an inseparability statement in weak form, because the shell parameter and the modular parameter are no longer independently adjustable.

4.3 Outer Standardness

The next observation is simple but essential: the ambient shells are standard for the same vector Ω. This is where the ambient realization is used as a regularity hypothesis rather than as extra dynamics.

Lemma 4.2 (Outer standardness)**.** *For every $\tau > 0$, the vector Ω is cyclic and separating for $\mathcal{A}(\tau)$.*

Proof. Since $\mathcal{A}_c \subset \mathcal{A}(\tau)$ and Ω is cyclic for $\mathcal{A}_c$ by $(U1)$, it is cyclic for $\mathcal{A}(\tau)$. Since $\mathcal{A}(\tau) \subset \mathcal{A}_U$ and Ω is separating for $\mathcal{A}_U$ by the ambient datum (Section 3.1), separation passes to the subalgebra $\mathcal{A}(\tau)$. □

This standardness will allow us to transport the outer shell family into an internal family inside $\mathcal{A}_c$ by Tomita-Takesaki duality.

4.4 The J-Dual Internal Filtration

The key construction is to turn the outer shell family into a family of subalgebras of the accessible algebra itself. This is the point at which the earlier interiorization axiom becomes unnecessary.

For $\mu > 0$, define

$$\mathcal{B}_\mu := J\,\mathcal{A}(1/\mu)'\,J.$$

The next proposition records the properties of this family needed for the HSMI step.

Proposition 4.3 (J-dual internal filtration)**.** *The family $\{\mathcal{B}_\mu\}_{\mu>0}$ satisfies:*

1. *$\mathcal{B}_\mu \subset \mathcal{A}_c$ for all $\mu > 0$;*
2. *if $\mu_1 > \mu_2$, then $\mathcal{B}_{\mu_1} \subset \mathcal{B}_{\mu_2}$;*
3. *Ω is cyclic and separating for each $\mathcal{B}_\mu$;*
4. *the modular group rescales the family according to*

$$\sigma_s(\mathcal{B}_\mu) = \mathcal{B}_{e^{-2\pi s}\mu};$$

5. *the family generates the accessible algebra:*

$$W^*\Big(\bigcup_{\mu>0}\mathcal{B}_\mu\Big)=\mathcal{A}_c.$$

Proof. The inclusion $\mathcal{A}_c\subset\mathcal{A}(1/\mu)$ implies

$$\mathcal{A}(1/\mu)'\subset\mathcal{A}_c',$$

hence

$$\mathcal{B}_\mu=J\,\mathcal{A}(1/\mu)'\,J\subset J\,\mathcal{A}_c'\,J=\mathcal{A}_c,$$

which proves (1).

For (2), if $\mu_1>\mu_2$, then $1/\mu_1<1/\mu_2$, so the monotonicity of the outer shell family gives

$$\mathcal{A}(1/\mu_1)\supset\mathcal{A}(1/\mu_2),$$

hence

$$\mathcal{A}(1/\mu_1)'\subset\mathcal{A}(1/\mu_2)'$$

and therefore

$$\mathcal{B}_{\mu_1}\subset\mathcal{B}_{\mu_2}.$$

For (3), Lemma 4.2 shows that Ω is cyclic and separating for $\mathcal{A}(1/\mu)$. Therefore Ω is also cyclic and separating for $\mathcal{A}(1/\mu)'$. Since $J\Omega=\Omega$ and J is antiunitary, the same holds for

$$J\,\mathcal{A}(1/\mu)'\,J=\mathcal{B}_\mu.$$

For (4), using the Tomita relation

$$\Delta^{is}J=J\Delta^{-is},$$

we compute

$$\sigma_s(\mathcal{B}_\mu)=\Delta^{is}J\,\mathcal{A}(1/\mu)'\,J\Delta^{-is}=J\Delta^{-is}\mathcal{A}(1/\mu)'\Delta^{is}J=J\,\sigma_{-s}\big(\mathcal{A}(1/\mu)\big)'\,J.$$

Now Proposition 4.1 gives

$$\sigma_{-s}\big(\mathcal{A}(1/\mu)\big)=\mathcal{A}(e^{2\pi s}/\mu),$$

so

$$\sigma_s(\mathcal{B}_\mu)=J\,\mathcal{A}(e^{2\pi s}/\mu)'\,J=\mathcal{B}_{e^{-2\pi s}\mu}.$$

Finally, (5) follows from the core limit in (Amb):

$$\mathcal{A}_c=\bigcap_{\tau>0}\mathcal{A}(\tau)\quad\Longrightarrow\quad\mathcal{A}_c'=W^*\Big(\bigcup_{\tau>0}\mathcal{A}(\tau)'\Big).$$

Applying $J(\cdot)J$ yields

$$\mathcal{A}_c=J\mathcal{A}_c'J=W^*\Big(\bigcup_{\tau>0}J\,\mathcal{A}(\tau)'\,J\Big)=W^*\Big(\bigcup_{\mu>0}\mathcal{B}_\mu\Big).$$

□

This construction is the first genuinely new structural step of the Phase 1 chain. The internal family is not imposed from outside. It is obtained functorially from the shell realization and modular data.

4.5 Half-Sided Modular Inclusion as a Theorem

We may now isolate the fixed inclusion that drives the Wiesbrock machinery.

Theorem 4.4 (HSMI from modular-decoherence covariance)**.** *For any fixed $\mu_0 > 0$, the inclusion*

$$(\mathcal{B}_{\mu_0} \subset \mathcal{A}_c, \Omega)$$

is a half-sided modular inclusion, namely

$$\sigma_t(\mathcal{B}_{\mu_0}) \subset \mathcal{B}_{\mu_0}, \qquad \forall t \leq 0.$$

Proof. By Proposition 4.3(3), Ω is cyclic and separating for both $\mathcal{B}_{\mu_0}$ and $\mathcal{A}_c$. By Proposition 4.3(4),

$$\sigma_t(\mathcal{B}_{\mu_0}) = \mathcal{B}_{e^{-2\pi t}\mu_0}.$$

If $t \leq 0$, then $e^{-2\pi t} \geq 1$, hence $e^{-2\pi t}\mu_0 \geq \mu_0$. By Proposition 4.3(2),

$$\mathcal{B}_{e^{-2\pi t}\mu_0} \subset \mathcal{B}_{\mu_0}.$$

Therefore

$$\sigma_t(\mathcal{B}_{\mu_0}) \subset \mathcal{B}_{\mu_0}, \qquad t \leq 0.$$

□

The important point is conceptual as much as technical: in the present framework, HSMI is not a geometric input imported from AQFT. It is the algebraic consequence of a scaling law that ties modular flow to decoherence depth.

4.6 The Wiesbrock-Borchers Positive Generator

Once the fixed inclusion is available, the standard Wiesbrock-Borchers conclusion follows.

Corollary 4.5 (Positive translation generator for the fixed inclusion)**.** *For each fixed $\mu_0 > 0$, there exists a unique positive self-adjoint operator*

$$P_{\mu_0} \geq 0$$

such that, with

$$U_{\mu_0}(a) := e^{-iaP_{\mu_0}}, \qquad a \geq 0,$$

one has

1. $U_{\mu_0}(a)\Omega = \Omega$ *for all* $a \geq 0$*;*
2. *the Borchers covariance relation*

$$\Delta_{\mathcal{A}_c}^{is} U_{\mu_0}(a) \Delta_{\mathcal{A}_c}^{-is} = U_{\mu_0}(e^{-2\pi s}a)$$

holds for all $s \in \mathbb{R}$ *and* $a \geq 0$.

Proof. This is the standard Wiesbrock theorem for a fixed half-sided modular inclusion [17], together with Borchers covariance [2]. □

At this stage we do not claim a pointwise identification between the entire family $\{\mathcal{B}_\mu\}_{\mu>0}$ and the Wiesbrock orbit generated by P_{μ_0}. Such an identification requires an additional normalization or uniqueness argument and is not needed for the Phase 1 main chain.

This completes the first half of Phase 1. We have shown that the accessible algebra canonically carries the half-sided modular structure needed to enter the Möbius reconstruction theorem.

4.7 GLW Conformal Skeleton and Weak Inseparability

The second half of Phase 1 upgrades the fixed half-sided modular inclusion to a full conformal skeleton. The only additional input is an existential conormality condition. Once this is imposed, the Guido-Longo-Wiesbrock reconstruction theorem produces a canonical strongly additive local Möbius net on S^1, and with it the conformal-level version of inseparability.

Throughout this section we assume, in addition to the hypotheses of Section 4.1, the conormality condition $(CN_\exists)$: there exists $\mu_0 > 0$ such that the vector Ω is cyclic for

$$\mathcal{C}_{\mu_0} := \mathcal{B}'_{\mu_0} \cap \mathcal{A}_c.$$

4.8 The Second Direction from Existential Conormality

The role of $(CN_\exists)$ is very precise. It is not used to construct the half-sided inclusion itself. That has already been achieved in Section 4.1. Rather, it upgrades the fixed inclusion to a standard one in the sense required by the GLW theorem.

For $\mu > 0$, define

$$\mathcal{C}_\mu := \mathcal{B}'_\mu \cap \mathcal{A}_c.$$

Proposition 4.6 (The second direction)**.** *Under $(U1)$, (C), (Amb), and $(CN_\exists)$, the family $\{\mathcal{C}_\mu\}_{\mu>0}$ satisfies:*

1. *$\mathcal{C}_\mu \subset \mathcal{A}_c$ and Ω is separating for $\mathcal{C}_\mu$;*
2. *the modular group rescales the family by*
$$\sigma_t(\mathcal{C}_\mu) = \mathcal{C}_{e^{-2\pi t}\mu};$$
3. *for all $t \geq 0$,*
$$\sigma_t(\mathcal{C}_\mu) \subset \mathcal{C}_\mu;$$
4. *if Ω is cyclic for one $\mathcal{C}_{\mu_0}$, then by modular propagation it is cyclic for all $\mathcal{C}_\mu$.*

In particular, the fixed inclusion

$$(\mathcal{B}_{\mu_0} \subset \mathcal{A}_c, \Omega)$$

is standard.

Proof. The inclusion $\mathcal{C}_\mu \subset \mathcal{A}_c$ is immediate from the definition. Separation of Ω follows from $(U1)$, since $\omega|_{\mathcal{A}_c}$ is faithful and faithfulness passes to subalgebras.

For modular covariance,

$$\sigma_t(\mathcal{C}_\mu) = \sigma_t(\mathcal{B}'_\mu \cap \mathcal{A}_c) = \sigma_t(\mathcal{B}_\mu)' \cap \sigma_t(\mathcal{A}_c) = \mathcal{B}'_{e^{-2\pi t}\mu} \cap \mathcal{A}_c = \mathcal{C}_{e^{-2\pi t}\mu},$$

where we used Proposition 4.3(4) and the tautological invariance of $\mathcal{A}_c$ under its own modular group.

If $t \geq 0$, then $e^{-2\pi t} \leq 1$, so the monotonicity of the $\mathcal{B}_\mu$ family implies the reverse monotonicity of the relative commutants:

$$\mathcal{C}_{e^{-2\pi t}\mu} \subset \mathcal{C}_\mu.$$

Thus

$$\sigma_t(\mathcal{C}_\mu) \subset \mathcal{C}_\mu.$$

Finally, $(CN_\exists)$ gives cyclicity for one $\mathcal{C}_{\mu_0}$, and the modular covariance formula propagates this cyclicity along the entire orbit. Since Section 4.1 already gives cyclicity and separation for $\mathcal{B}_{\mu_0}$ and $\mathcal{A}_c$, the inclusion is standard. □

This is the exact point at which the second direction closes. No stronger global conormality assumption is needed for the fixed inclusion that feeds the GLW theorem.

4.9 The GLW Reconstruction Theorem

Once standardness is available, the conformal skeleton appears by a known AQFT theorem. The novelty in the present paper is not the GLW theorem itself, but the conditional construction of its input from the explicit Paper I shell interface and dynamical package.

Theorem 4.7 (Conformal skeleton via Guido-Longo-Wiesbrock)**.** *Fix $\mu_0 > 0$ as in $(CN_\exists)$. Then the standard half-sided modular inclusion*

$$(\mathcal{B}_{\mu_0} \subset \mathcal{A}_c, \Omega)$$

determines a unique strongly additive local Möbius covariant net

$$I \longmapsto \mathcal{A}_{\mathrm{GLW}}(I)$$

on S^1 such that

$$\mathcal{A}_{\mathrm{GLW}}(0, +\infty) = \mathcal{A}_c, \qquad \mathcal{A}_{\mathrm{GLW}}(1, +\infty) = \mathcal{B}_{\mu_0},$$

and Ω is the vacuum vector for the corresponding positive-energy representation of $\mathrm{PSL}(2, \mathbb{R})$.

Proof. This is precisely the Guido-Longo-Wiesbrock reconstruction theorem for standard half-sided modular inclusions [5]. □

Two clarifications are important here.

First, at this stage no separate relative-position hypothesis is needed. Standard HSMI is enough to reconstruct the Möbius net. The previously suspected relative-position bottleneck concerns later identification questions, not the existence of the conformal skeleton itself.

Second, the net produced here is an abstract but canonical conformal object attached to the fixed inclusion. The remaining question of identifying this GLW net pointwise with the pre-given shell filtrations or later cross-ratio families belongs to the next phase of the program.

4.10 The Conformal Skeleton as the Output of Modular Data

Theorem 4.7 already yields the first inseparability statement of the paper. The data entering the standard inclusion are modular data; the output is a conformal geometry in the minimal sense appropriate to one spatial and one temporal dimension, namely a causal/conformal skeleton.

To make this precise, let

$$\mathrm{Mod}_{\mathrm{conf}}(\mathcal{A}_c, \omega)$$

denote the conformal modular sector determined in Phase 1: the modular group of $(\mathcal{A}_c, \Omega)$, the fixed-inclusion positive generator P_{μ_0}, and the corresponding Möbius representation recovered from the standard HSMI. Let

$$\mathrm{Conf}(\mathcal{A}_c, \omega)$$

denote the conformal skeleton, equivalently the GLW net together with its Möbius vacuum representation.

Theorem 4.8 (Conditional Weak Inseparability)**.** *Conditional on the supplied shell interface (Amb), the independent dynamical axiom (C), $(U1)$, $(CN_\exists)$, and the carrier condition $(U3)$, the conformal skeleton and the conformal modular sector cannot be independently varied. More precisely:*

1. *the standard HSMI data determine a unique conformal skeleton;*
2. *the conformal skeleton, once normalized by the half-line identification above, determines the conformal modular sector;*
3. *therefore any admissible deformation that preserves one of these two structures must preserve the other.*

Proof. Item (1) is exactly Theorem 4.7.

For item (2), the Möbius net comes equipped with a positive-energy representation of $\mathrm{PSL}(2,\mathbb{R})$. Once the half-line normalization

$$\mathcal{A}_{\mathrm{GLW}}(0,+\infty) = \mathcal{A}_c, \qquad \mathcal{A}_{\mathrm{GLW}}(1,+\infty) = \mathcal{B}_{\mu_0}$$

is fixed, the corresponding dilation and translation subgroups recover the modular action and the positive translation generator associated with the inclusion. In other words, the conformal skeleton rigidly determines the conformal modular sector.

Item (3) is immediate. If an admissible deformation preserves the conformal modular sector, item (1) forces the conformal skeleton to remain fixed. Conversely, if it preserves the conformal skeleton, item (2) forces the modular sector to remain fixed. Hence the two structures are not independently deformable. □

This result is weak only in the sense that it stops at the conformal level. It does not yet recover metric scale. That is the purpose of Step I and Step II. But it already establishes the central conceptual point of Phase 1:

> causal/conformal geometry and modular time are not two separately specified ingredients;
> they are two readings of the same standard half-sided modular data.

4.11 What Phase 1 Has Actually Closed

It is useful to isolate the precise mathematical output of the section before moving on to metric scale.

1. HSMI is not separately assumed. It is a conditional theorem derived from the explicit interface (Amb) and dynamical axiom (C).
2. The second direction is no longer a global mystery. It is isolated into the existence-type conormality input $(CN_{\exists})$.
3. The conformal skeleton is no longer conjectural. It is reconstructed canonically by GLW from the fixed standard inclusion.
4. The first inseparability theorem is already in hand: conformal geometry and conformal modular data are mutually rigid.

What remains open after Phase 1 is not the existence of the conformal skeleton, but the recovery of metric scale from local modular data. That is why the next phase turns to the cross-ratio family and to the canonical local modular scale datum

$$\{K_b\}_{b\in(0,1)}.$$

5 From Conformal Skeleton to Full Geometry (Phase 2)

Phase 1 established that conformal geometry and conformal modular data are inseparable. But the conformal skeleton determines only the causal structure, the light cones. It does not determine distances. A conformal class of Lorentzian metrics contains infinitely many metrics related by Weyl rescaling. To reach the full inseparability statement, we must show that the metric scale is also locked to modular data.

This is the task of Phase 2. The strategy has three steps: identify the correct local geometric modulus, extract a canonical modular scale datum from it, and show that this datum determines the metric scale via a Type III-compatible quantum Fisher information metric.

5.1 Why Single-Interval Deformations Fail

The first instinct is to deform a single interval within the GLW net and track how the modular Hamiltonian changes. This fails for a precise reason.

On S^1, the group $\mathrm{PSL}(2,\mathbb{R})$ acts transitively on the space of intervals. Any single-interval deformation, such as moving one endpoint while fixing the other, is a Möbius transformation. Since the vacuum state ω is Möbius-invariant, pulling back the restricted state along such a deformation produces a constant family. The quantum Fisher information metric of a constant family is zero.

The lesson is that single-interval endpoint motion is a gauge orbit, not a geometric deformation. To obtain a nontrivial modulus, one needs at least four boundary points.

5.2 The Cross-Ratio as the Minimal Non-Möbius Modulus

Consider a nested double-interval configuration within the GLW net:

$$\mathcal{M}_{\text{in}}(b) \subset \mathcal{M}_{\text{out}}, \qquad b \in (0,1).$$

In the standard half-line gauge, this means

$$\mathcal{M}_{\text{out}} = \mathcal{A}_{\text{GLW}}(0,\infty), \qquad \mathcal{M}_{\text{in}}(b) = \mathcal{A}_{\text{GLW}}(b,1).$$

The four boundary points are $0, b, 1, \infty$. The Möbius group has three real parameters, so after gauge-fixing, exactly one real modulus survives: the cross-ratio

$$\chi = b.$$

This is the minimal non-Möbius geometric degree of freedom available in the GLW net. Varying b genuinely changes the relative position of the inner and outer algebras in a way that no Möbius transformation can undo.

5.3 The Canonical Modular Scale Datum

For each $b \in (0,1)$, the inclusion $\mathcal{M}_{\text{in}}(b) \subset \mathcal{M}_{\text{out}}$ determines a modular operator $\Delta_{\mathcal{M}_{\text{in}}(b),\Omega}$ via the Tomita-Takesaki theorem applied to $\mathcal{M}_{\text{in}}(b)$ with respect to the vacuum vector Ω.

Definition 5.1 (Canonical modular scale datum)**.**

$$K_b := -\ln \Delta_{\mathcal{M}_{\text{in}}(b),\Omega}, \qquad b \in (0,1).$$

The family $\{K_b\}_{b\in(0,1)}$ is the canonical local modular scale datum. It is determined entirely by the GLW net and the vacuum state. No additional choices are involved.

Proposition 5.2 (Cross-ratio is the minimal non-Möbius modulus)**.** *A nested double-interval configuration in the GLW net has exactly one Möbius-invariant modulus, the cross-ratio $\chi = b$. In particular, the family $\{K_b\}$ captures the minimal local modular scale information that is not already contained in the conformal skeleton.*

This is the theorem-level output of Step I. It requires no additional axioms beyond Phase 1.

5.4 The Chart Axiom and the Scale Shadow

To convert the modular scale datum into a Riemannian metric on the cross-ratio parameter space, we need to differentiate the family $\{K_b\}$ and evaluate a quadratic form. This requires regularity that is not automatic from the abstract GLW construction.

Axiom (LImp^1). For each reference point $b_0 \in (0,1)$, there exists a neighborhood U_{b_0} and a unitary family

$$W_b \in \mathcal{U}(\mathcal{M}_{\text{out}}), \qquad b \in U_{b_0},$$

satisfying:

1. $W_{b_0} = \mathbf{1}$;
2. $W_b \mathcal{M}_* W_b^* = \mathcal{M}_{\text{in}}(b)$ for a fixed reference algebra $\mathcal{M}_* = \mathcal{M}_{\text{in}}(b_0)$;
3. $b \mapsto \omega(W_b A W_b^*)$ is C^1 for each $A \in \mathcal{M}_*$;
4. the tangent vector $\dot{\phi}_{b_0}$ at $b = b_0$ is represented by the modular Hamiltonian variation $\dot{K}_{b_0} = \frac{d}{db}|_{b=b_0} K_b$ via the standard Araki correspondence.

This axiom packages three regularity conditions into one: the existence of a local implementer (CR1), C^1 smoothness of the pulled-back state family (CR2), and tangent compatibility with the modular Hamiltonian variation (CR3).

Axiom (CR-nd) (Non-degeneracy). The Araki-Hessian quadratic form $\mathbf{Q}_{\omega,b}$ is strictly positive on the modular Hamiltonian variation:

$$\mathbf{Q}_{\omega,b}(\partial_b K_b, \partial_b K_b) > 0, \qquad \forall\, b \in (0,1).$$

In the worked examples, this follows from the positivity of the stress-tensor two-point function. It fails only if the cross-ratio deformation produces no state-level change, a physically degenerate situation.

Under these two conditions, the scale shadow is a genuine Riemannian metric on the cross-ratio parameter space.

Theorem 5.3 (Scale shadow, conditional on (LImp^1) + (CR-nd))**.** *Under the Phase 1 axioms and the additional conditions $(LImp^1)$ and $(CR\text{-}nd)$, the canonical modular scale datum induces a Riemannian scale shadow*

$$h_b = \mathbf{Q}_{\omega,b}(\partial_b K_b, \partial_b K_b)\, db^2,$$

where $\mathbf{Q}_{\omega,b}$ is the positive quadratic form induced by the Araki relative entropy Hessian [1] at the state $\omega|_{\mathcal{M}_{\text{in}}(b)}$ via the $(LImp^1)$ chart.

Remark 5.4. In the special case where the chart is of exponential type (CR-exp), this reduces to the familiar variance formula

$$h_b = \text{Var}_\omega(\partial_b K_b)\, db^2.$$

The general formula via $\mathbf{Q}_{\omega,b}$ does not require this specialization.

5.5 Geometric Datum: Conformal Skeleton + Slice Scale

The conformal skeleton from Phase 1 determines the causal structure, equivalently the conformal causal structure. The scale shadow $\{h_b\}$ determines the complementary scale information along the cross-ratio slice family. At the abstract level established in this paper, the pair $(\mathrm{Conf}, \{h_b\})$ is the full geometric datum. In the worked examples of Section 6, the cross-ratio parameter b naturally parametrizes the spatial position of the inner interval endpoint within the outer interval. The scale shadow h_b therefore directly defines a Riemannian metric on the spatial slice, and together with the conformal causal structure determines a Lorentzian metric.

Corollary 5.5 (Full geometric datum, 1+1 dimensions)**.** *In the 1+1 dimensional setting of the GLW net, the conformal skeleton* $\mathrm{Conf}(\mathcal{N}, \omega)$ *together with the scale family* $\{h_b\}_{b\in(0,1)}$ *defines the full geometric datum*

$$\mathrm{Geom}_{\mathrm{full}}(\mathcal{N}, \omega) := \big(\mathrm{Conf}(\mathcal{N}, \omega),\ \{h_b\}_{b\in(0,1)}\big).$$

In the worked examples of Section 6, the cross-ratio parameter b naturally parametrizes the spatial position of the inner interval endpoint within the outer interval. The scale shadow h_b therefore directly defines a Riemannian metric on the spatial slice, and together with the conformal causal structure determines a Lorentzian metric.

Remark 5.6. The step from scale shadow to Lorentzian metric relies on the model-level identification of the cross-ratio parameter with spatial position. At the abstract level, Corollary 5.5 provides a full geometric datum; the Lorentzian interpretation is verified in the worked examples but not proved in full generality. The extension to higher dimensions is an open problem (Section 10.2).

5.6 Forward and Bidirectional Full Inseparability

With the full geometric datum in hand, we can state the main theorems.

Definition 5.7 (Extended modular data)**.**

$$\mathrm{Mod}_{\mathrm{ext}}(\mathcal{N}, \omega) := \big(\mathrm{Mod}_{\mathrm{conf}}(\mathcal{N}, \omega),\ \{K_b\}_{b\in(0,1)}\big).$$

Two extended modular data $\mathrm{Mod}^{(1)}_{\mathrm{ext}}$ and $\mathrm{Mod}^{(2)}_{\mathrm{ext}}$ are identified if their conformal modular sectors agree and their canonical scale families $\{K_b\}$ agree up to the trivial equivalences specified in (SFa). All equalities of $\mathrm{Mod}_{\mathrm{ext}}$ below are understood in this quotient sense.

Theorem 5.8 (Forward Full Inseparability)**.** *Under the Phase 1 axioms and* $(LImp^1)$ + $(CR\text{-}nd)$*, if two admissible objects have the same extended modular data, then they have the same full geometric datum:*

$$\mathrm{Mod}^{(1)}_{\mathrm{ext}} = \mathrm{Mod}^{(2)}_{\mathrm{ext}} \quad \Longrightarrow \quad \mathrm{Geom}^{(1)}_{\mathrm{full}} = \mathrm{Geom}^{(2)}_{\mathrm{full}}.$$

Proof. By Phase 1 Weak Inseparability, the conformal modular data determine the conformal skeleton. By Theorem 5.3, the canonical modular scale family $\{K_b\}$ determines the scale shadow $\{h_b\}$. By Corollary 5.4, the conformal skeleton and scale shadow together determine the full geometric datum. □

The forward direction says: you cannot change the full geometric datum without changing the modular data. But can you change the modular data without changing the full geometric datum? This requires an additional faithfulness condition.

Axiom (SFa) (Scale faithfulness). The canonical scale map $\{K_b\} \mapsto \{h_b\}$ is faithful within the admissible class: if $h_b^{(1)} = h_b^{(2)}$ for all b, then $K_b^{(1)} = K_b^{(2)}$ for all b, up to trivial equivalences that do not affect the scale map.

Theorem 5.9 (Forward result and conditional converse)**.** *Under the Phase 1 axioms,* $(LImp^1)$*, and* $(CR\text{-}nd)$*, the derived implication is*

$$\mathrm{Mod}_{\mathrm{ext}}^{(1)} = \mathrm{Mod}_{\mathrm{ext}}^{(2)} \quad \Longrightarrow \quad \mathrm{Geom}_{\mathrm{full}}^{(1)} = \mathrm{Geom}_{\mathrm{full}}^{(2)}.$$

If the additional scale-faithfulness axiom (SFa) *is supplied, then the converse implication also holds, and hence*

$$\mathrm{Geom}_{\mathrm{full}}^{(1)} = \mathrm{Geom}_{\mathrm{full}}^{(2)} \quad \Longleftrightarrow \quad \mathrm{Mod}_{\mathrm{ext}}^{(1)} = \mathrm{Mod}_{\mathrm{ext}}^{(2)}.$$

Proof. The displayed forward implication is Theorem 5.6. For the conditional converse: by the definition of $\mathrm{Geom}_{\mathrm{full}}$, equality of full geometries determines both the conformal skeleton and the scale family $\{h_b\}$. The conformal skeleton determines the conformal modular data by Phase 1 Weak Inseparability. The scale family determines the canonical modular scale family $\{K_b\}$ by (SFa). Together, $\mathrm{Mod}_{\mathrm{ext}}^{(1)} = \mathrm{Mod}_{\mathrm{ext}}^{(2)}$. □

The converse is therefore not an independently derived theorem: its injective step is exactly the model-level axiom (SFa). Under the identifications defended in Section 2, the mathematical counterpart of physical structure (the full geometric datum $\mathrm{Geom}_{\mathrm{full}}$) and the mathematical counterpart of algebraic modular time (extended modular data) cannot be independently specified within this conditional package. Modular flow supplies an algebraic time parameter only. Identifying that parameter with phenomenal or first-person time is an additional interpretive move, not a conclusion of Theorems 4.8 or 5.9.

6 Abstract Sufficient Conditions for $(LImp^1)$

The Phase 2 theorems are conditional on the chart axiom $(LImp^1)$, the non-degeneracy condition $(CR\text{-}nd)$, and the faithfulness condition (SFa). Of these three, $(LImp^1)$ is the most structurally demanding: it requires a local unitary implementer family with C^1 regularity and tangent compatibility. This section identifies two abstract routes for discharging $(LImp^1)$: a full discharge for Diff-covariant nets, and a decomposition of the remaining non-Diff burden into transportability, inner implementability, and regularity.

The conditions $(CR\text{-}nd)$ and (SFa) are not addressed here. They appear to depend on dynamical and spectral properties of the specific model (e.g., the positivity of the stress-tensor two-point function for $(CR\text{-}nd)$, and the injectivity of the variance map for (SFa)), and are better treated as model-level verifications rather than consequences of structural regularity.

6.1 Decomposition of $(LImp^1)$

It is useful to isolate the three independent requirements packaged into $(LImp^1)$:

1. **Local transport.** For each b near a reference point b_0, there exists an isomorphism $\alpha_b \colon \mathcal{M}_* \to \mathcal{M}_{\mathrm{in}}(b)$ that extends to an automorphism of $\mathcal{M}_{\mathrm{out}}$.

2. **Unitary implementability.** The automorphism α_b is inner in $\mathcal{M}_{\text{out}}$: there exists $W_b \in \mathcal{U}(\mathcal{M}_{\text{out}})$ with $\alpha_b = \text{Ad}(W_b)$.

3. C^1 **regularity and tangent compatibility.** The map $b \mapsto \omega(W_b A W_b^*)$ is C^1 for each $A \in \mathcal{M}_*$, and the tangent at b_0 is represented by the modular Hamiltonian variation $\dot{K}_{b_0}$ via the Araki correspondence.

Different sufficient conditions address different subsets of these requirements.

6.2 Why Split Property Alone Is Not Sufficient

The split property for the inclusion $\mathcal{M}_{\text{in}}(b) \subset \mathcal{M}_{\text{out}}$ guarantees the existence of a Type I factor $\mathcal{N}$ interpolating between the two algebras, and hence a local tensor product decomposition of the Hilbert space. This is relevant to requirement (2): in the presence of a split, the "annular region" between the inner and outer algebras has enough room to localize unitary implementers.

However, the split property does not by itself produce the transport of requirement (1). It tells you that the relative position of $\mathcal{M}_{\text{in}}(b)$ inside $\mathcal{M}_{\text{out}}$ is "standard" in a measure-theoretic sense, but it does not provide a canonical family of automorphisms mapping $\mathcal{M}_*$ to $\mathcal{M}_{\text{in}}(b)$ as b varies. That family must come from an external source: either a symmetry group acting on the net, or an explicit transportability assumption.

Nor does the split property address requirement (3). The C^1 regularity of the pulled-back state family and its tangent compatibility with the modular Hamiltonian variation are analytic conditions that depend on the smoothness of whatever generates the transport, not on the split structure.

In summary: split property assists with unitary implementability but does not generate transport or regularity. It is a facilitating condition, not a sufficient one.

6.3 Diff-Covariant Nets

The cleanest sufficient condition for $(LImp^1)$ is diffeomorphism covariance of the net.

Theorem 6.1 (Diff-covariance implies $(LImp^1)$)**.** *Let $I \mapsto \mathcal{A}(I)$ be a local net on S^1 (or on the half-line) that is covariant under a strongly continuous positive-energy representation of* Diff(S^1) *(respectively, of orientation-preserving diffeomorphisms of the half-line fixing the boundary). Then $(LImp^1)$ holds for the cross-ratio family $\{\mathcal{M}_{\text{in}}(b)\}_{b\in(0,1)}$ inside $\mathcal{M}_{\text{out}}$.*

Proof. Fix a reference point $b_0 \in (0,1)$ and set $\mathcal{M}_* = \mathcal{M}_{\text{in}}(b_0)$. For each b in a neighborhood of b_0, choose a smooth family of diffeomorphisms f_b satisfying $f_{b_0} = \text{id}$, $f_b(b_0) = b$, $f_b(1) = 1$, with support contained in the outer interval.

Transport. The diffeomorphism covariance of the net gives $f_b(\mathcal{M}_*) = \mathcal{M}_{\text{in}}(b)$, and since $\text{supp}(f_b)$ is contained in the outer interval, the corresponding automorphism α_b restricts to an automorphism of $\mathcal{M}_{\text{out}}$. This establishes requirement (1).

Unitary implementability. The positive-energy representation provides unitaries $U(f_b)$ implementing α_b. Since f_b is supported in the outer interval, $U(f_b) \in \mathcal{M}_{\text{out}}$ by Haag duality (or by the localization properties of the Diff representation). Set $W_b := U(f_b)$. This establishes requirement (2).

C^1 regularity. The map $b \mapsto f_b$ is smooth by construction. The strong continuity of the Diff representation ensures that $b \mapsto U(f_b)$ is strongly continuous, and the smoothness of f_b upgrades this to C^1 regularity of the pulled-back state family $b \mapsto \omega(W_b A W_b^*)$.

Tangent compatibility. The infinitesimal generator of the family $U(f_b)$ at $b = b_0$ is a smeared stress-energy tensor supported in the outer interval. By the relation between stress-energy deformations and modular Hamiltonian variations in conformal field theory [3], this generator represents the modular Hamiltonian variation $\dot{K}_{b_0}$ via the Araki correspondence. This establishes requirement (3).

Nontriviality. The diffeomorphisms f_b are not Möbius transformations (they fix boundary points that a Möbius transformation would move). Therefore the pulled-back state family is nonconstant, and the QFIM is nonzero. □

This theorem upgrades the Virasoro worked example of Section 7.2 from a model verification to an instance of a general result: any Diff-covariant conformal net automatically satisfies $(LImp^1)$.

6.4 Transportability-Based Sufficiency

For nets that are not Diff-covariant, a second sufficient condition can be formulated by combining an explicit transportability hypothesis with the injectivity of the outer algebra.

Definition 6.2 (Local transportability)**.** The cross-ratio family $\{\mathcal{M}_{\text{in}}(b)\}$ is *locally transportable* inside $\mathcal{M}_{\text{out}}$ if, for each $b_0 \in (0,1)$, there exists a neighborhood U_{b_0} and a family of automorphisms $\alpha_b \in \text{Aut}(\mathcal{M}_{\text{out}})$, $b \in U_{b_0}$, with $\alpha_{b_0} = \text{id}$ and $\alpha_b(\mathcal{M}_*) = \mathcal{M}_{\text{in}}(b)$.

Theorem 6.3 (Transportability-based sufficiency for $(LImp^1)$)**.** *Suppose:*

1. *the cross-ratio family is locally transportable inside $\mathcal{M}_{\text{out}}$;*
2. *$\mathcal{M}_{\text{out}}$ is an injective (equivalently, hyperfinite) Type III$_1$ factor;*
3. *the transport family $b \mapsto \alpha_b$ is C^1 in the sense that $b \mapsto \omega(\alpha_b(A))$ is C^1 for each $A \in \mathcal{M}_*$, with tangent compatible with the modular Hamiltonian variation.*

Then $(LImp^1)$ holds.

Proof. Requirement (1) of $(LImp^1)$ is exactly hypothesis (1). For requirement (2): by hypothesis (2) and the Connes theorem [14] ($\text{Out}(\mathcal{M}) = \{e\}$ for injective Type III$_1$ factors), every automorphism of $\mathcal{M}_{\text{out}}$ is inner. Hence $\alpha_b = \text{Ad}(W_b)$ for some $W_b \in \mathcal{U}(\mathcal{M}_{\text{out}})$. Requirement (3) is hypothesis (3). □

Remark 6.4. If the inclusion $\mathcal{M}_{\text{in}}(b) \subset \mathcal{M}_{\text{out}}$ additionally satisfies the split property, then the Type I interpolation allows W_b to be chosen with support localized in the annular region between the two algebras. This localization can improve the analytic control needed for the C^1 regularity of hypothesis (3), but it is not required for the theorem as stated.

It is important to be clear about what this theorem does and does not achieve. Hypotheses (1) and (3) —transportability and C^1 tangent compatibility —are already the two main substantive components of $(LImp^1)$ itself. The theorem's contribution is to identify the precise algebraic condition (injectivity of the outer factor) that bridges the gap between "transport exists as an automorphism" and "transport is unitarily implemented inside the outer algebra." For nets whose local algebras are known to be injective Type III$_1$ factors —which includes all local algebras in standard AQFT models —this bridge is automatic.

6.5 Status of the Three Phase 2 Conditions

The results of this section change the status of $(LImp^1)$ as follows:

- For Diff-covariant nets, $(LImp^1)$ is fully discharged by Theorem 6.1. No additional assumptions are needed.
- For non-Diff-covariant nets with injective Type III_1 local algebras, $(LImp^1)$ reduces to local transportability plus C^1 regularity (Theorem 6.3). The remaining burden is to verify that the cross-ratio family admits a smooth transport, which is a geometric property of the net rather than an abstract axiom.
- $(CR\text{-}nd)$ and (SFa) remain model-level verifications. They depend on dynamical and spectral properties (positivity of the stress-tensor two-point function, injectivity of the scale map) that are not captured by structural regularity conditions.

The net effect is that, for Diff-covariant nets, the Phase 2 conditionality is reduced from three independent axioms to two model-level verifications. For non-Diff-covariant nets, the situation is more modest: $(LImp^1)$ is decomposed and partially discharged, but the transportability and regularity inputs remain substantive.

7 Verification: Two Worked Examples

The abstract Phase 2 result is deliberately conditional. Of the three conditions, $(LImp^1)$ now has abstract sufficient conditions (Section 6): it is fully discharged for Diff-covariant nets, and for other nets the remaining burden is isolated to transportability and regularity. The remaining conditions $(CR\text{-}nd)$ and (SFa) are verified in concrete models. This section provides that verification in two structurally distinct examples.

The examples are chosen to do different kinds of work. The Rindler/free-field chiral half-line model shows that the Phase 2 package is realized in the same physical family that motivated much of the original HAFF intuition. The Virasoro-net example then shows that the construction is not a free-field accident and, in particular, that the chart axiom becomes especially natural once $\mathrm{Diff}(S^1)$ covariance is built in.

7.1 Worked Example I: Rindler / Free-Field Chiral Half-Line

The first example must be formulated carefully. The correct Phase 2 object is not the family of translated Rindler wedges. That family is useful as a sanity check for the Phase 1 half-sided modular structure, but it is the wrong object for the cross-ratio construction because global translations leave the vacuum invariant and thus produce a constant pulled-back state family. The correct model is the chiral half-line realization associated with the Rindler horizon.

We therefore take:

$$\mathcal{M}_{\text{out}} = \mathcal{A}(0,\infty), \qquad \mathcal{M}_{\text{in}}(b) = \mathcal{A}(b,1), \qquad b \in (0,1),$$

in the vacuum representation of a chiral free-field net. The canonical modular scale datum is then

$$K_b := -\ln \Delta_{\mathcal{M}_{\text{in}}(b),\Omega}.$$

7.1.1 Verification of $(LImp^1)$

In this model, the required local chart is supplied by the local diffeomorphism covariance of the free-field chiral net. Specifically, we assume covariance under orientation-preserving diffeomorphisms of the half-line that fix the boundary, implemented by unitaries in the corresponding local algebras. Fix a reference point $b_0 \in (0,1)$ and set

$$\mathcal{M}_* := \mathcal{A}(b_0, 1).$$

Choose a smooth local family of diffeomorphisms

$$f_b : (0,\infty) \to (0,\infty), \qquad b \in U_{b_0},$$

such that

1. $f_{b_0} = \mathrm{id}$;
2. $f_b(1) = 1$;
3. $f_b(b_0) = b$;
4. the support of f_b is contained in the outer half-line.

This covariance gives unitaries

$$W_b := U(f_b) \in \mathcal{U}(\mathcal{M}_{\text{out}})$$

with

$$W_b \mathcal{M}_* W_b^* = \mathcal{M}_{\text{in}}(b).$$

Because the deformation is genuinely boundary-fixed rather than Möbius, the family is nontrivial; because $b \mapsto f_b$ is smooth, the pulled-back state family is C^1; and because the infinitesimal generator is represented by the stress tensor, the tangent of the state family is compatible with the modular Hamiltonian variation. Thus $(LImp^1)$ holds in the model.

7.1.2 Verification of $(CR\text{-}nd)$

In a vacuum chiral conformal theory, the modular Hamiltonian of an interval (a,b) has the local CHM form [3]

$$K_{(a,b)} = 2\pi \int_a^b \frac{(x-a)(b-x)}{b-a}\, T_{00}(x)\, dx.$$

Specializing to the present family gives

$$K_b = 2\pi \int_b^1 \frac{(x-b)(1-x)}{1-b}\, T_{00}(x)\, dx,$$

and differentiation yields

$$\partial_b K_b = -2\pi \int_b^1 \frac{(1-x)^2}{(1-b)^2}\, T_{00}(x)\, dx.$$

The corresponding Araki-Hessian quadratic form is positive and nonzero, so the scale shadow is non-degenerate. More explicitly, one obtains

$$h_b = C_R (1-b)^{-2} db^2, \qquad C_R > 0.$$

Hence $(CR\text{-}nd)$ is satisfied.

7.1.3 Verification of (SFa)

In this model, faithfulness reduces to a one-dimensional injectivity statement. Since

$$h_b = C_R(1-b)^{-2}db^2$$

is strictly monotone in b, the metric profile determines the parameter b, and hence determines the corresponding modular datum K_b up to the trivial equivalences already allowed in the definition of (SFa). Therefore the scale map is faithful in the required sense.

The upshot is that the entire Phase 2 package is realized in the Rindler/free-field chiral half-line model:

$$(\text{LImp}^1), \quad (\text{CR-nd}), \quad (\text{SFa}).$$

7.2 Worked Example II: The Virasoro Net

The second example shows that the preceding mechanism is not tied to free fields. Let

$$I \longmapsto \mathcal{A}_{\text{Vir},c}(I)$$

be the Virasoro net at central charge c, with vacuum vector Ω. As in the first example, we work in the standard half-line gauge:

$$\mathcal{M}_{\text{out}} = \mathcal{A}_{\text{Vir},c}(0,\infty), \qquad \mathcal{M}_{\text{in}}(b) = \mathcal{A}_{\text{Vir},c}(b,1).$$

7.2.1 Verification of $(LImp^1)$

What was model-dependent in the first example becomes built in here. The Virasoro net is Diff(S^1)-covariant, so $(LImp^1)$ follows directly from Theorem 6.1. Concretely, one can choose a smooth boundary-fixed family

$$f_b \in \text{Diff}(S^1), \qquad b \in U_{b_0},$$

with

$$f_{b_0} = \text{id}, \qquad f_b(b_0) = b, \qquad f_b(1) = 1,$$

and support contained in the outer half-line. The positive-energy Diff(S^1) representation yields unitaries

$$W_b := U(f_b) \in \mathcal{U}(\mathcal{M}_{\text{out}})$$

such that

$$W_b \mathcal{A}(b_0,1) W_b^* = \mathcal{A}(b,1).$$

The nontriviality, smoothness, and tangent compatibility required by $(LImp^1)$ follow immediately from the non-Möbius character of f_b, the smoothness of the Diff(S^1) action, and the fact that the infinitesimal action is generated by the Virasoro stress tensor. Thus $(LImp^1)$ holds here in a cleaner and more canonical way than in the free-field model.

7.2.2 Verification of $(CR\text{-}nd)$

The interval modular Hamiltonian again has the CHM form

$$K_b = 2\pi \int_b^1 \frac{(x-b)(1-x)}{1-b}\, T(x)\, dx,$$

where T is now the Virasoro stress tensor. Differentiating gives the same structural formula as before:

$$\partial_b K_b = -2\pi \int_b^1 \frac{(1-x)^2}{(1-b)^2}\, T(x)\, dx.$$

Using the Virasoro two-point function

$$\langle T(x)T(y)\rangle \sim \frac{c/2}{(x-y)^4},$$

one obtains, after standard regularization, a positive finite coefficient

$$\mathbf{Q}_{\omega,b}(\partial_b K_b, \partial_b K_b) = \frac{C_{\text{Vir}}(c)}{(1-b)^2}, \qquad C_{\text{Vir}}(c) > 0.$$

Hence

$$h_b = C_{\text{Vir}}(c)(1-b)^{-2} db^2,$$

and $(CR\text{-}nd)$ holds.

7.2.3 Verification of (SFa)

Exactly as in the first example, the scale profile is strictly injective in the cross-ratio parameter. Therefore the full scale family determines b, and hence determines the local modular datum up to the same trivial equivalences. Thus (SFa) is satisfied in the Virasoro model as well.

7.3 Comparison and Universal Scaling

The significance of the two examples lies less in their individual computations than in what they share.

1. Both models realize the same abstract Phase 2 package:
$$(\text{LImp}^1) + (\text{CR-nd}) + (\text{SFa}).$$
2. Both produce the same scale profile:
$$h_b = C(1-b)^{-2} db^2.$$
3. The model dependence enters only through the coefficient:
   - in the free-field model through the normalization of the stress-tensor sector;
   - in the Virasoro model through the central charge c.

This is the strongest available evidence, within the present paper, that the scale shadow is not a model-specific artifact. What appears to be universal is the cross-ratio scaling law itself, which is fixed by the CHM structure of interval modular Hamiltonians. What remains model-sensitive is only the overall constant.

From the point of view of the paper's philosophical thesis, this matters because it shows that the full inseparability mechanism is not tied to one special solvable example. The rigidity claim is abstract; the examples show that the additional Phase 2 inputs are concretely realizable in more than one structural setting, and that the theorem's conditions arise naturally where one would most expect them to arise.

8 Implications for the Hard Problem

For the purposes of this section, we take the mathematical results of the previous sections as given. At the conformal level, Theorem 4.8 already shows that conformal geometry and conformal modular data are not independently deformable. The later Phase 2 results strengthen this to the full scale-sensitive level by tying the metric scale to the canonical modular scale datum. The philosophical question is therefore no longer whether one can attach a speculative interpretation to a formal analogy. It is whether the standard formulation of the hard problem survives once its key presupposition is written in mathematically precise terms.

8.1 The Failure of the Presupposition

Section 1 identified the hidden presupposition of the hard problem: independent describability. Section 2 translated it into independent deformability. The inseparability theorems of Sections 4 and 5 now supply the answer.

At the Phase 1 level, Theorem 4.8 shows that conformal geometry and conformal modular data are already locked together. At the full level, Theorem 5.7 shows that, under $(LImp^1)$, $(CR\text{-}nd)$, and (SFa), the full geometric datum and the extended modular data cannot be independently varied. Conditional on the identifications defended in Section 2, the presupposition is not available at either level.

The strongest claim justified by the theorem is therefore not that the hard problem has been "solved" in the ordinary sense. It is that its canonical production form is ill-posed. The question "how does matter produce mind?" asks for a derivation across a divide that, in the present framework, is not mathematically available as an independent divide. One cannot first specify the geometric-physical side and only afterwards ask how the first-person side appears. They are two structurally inseparable readings of the same algebraic datum $(\mathcal{A}, \omega)$.

8.2 Relocating the Explanatory Gap

Once the production picture is rejected, a residual question remains. Even if geometry and modular flow are inseparable, why should a given modular invariant correspond to a given phenomenal quality? This is legitimate, but it is not the original hard problem. It concerns the interpretive reach of an observer embedded in the very structure being described.

The proper location of the remaining gap is not between physical and experiential structure as two externally juxtaposed domains. A possible research direction places it within the self-description limits of a finite agent modeled by $(\mathcal{A}, \omega)$. Paper G's discussion of self-reference [9] motivates this direction, but it does not prove a generic fixed-point no-go or an exhaustive regress-versus-collapse dichotomy. No theorem of the present paper therefore infers a phenomenal limitation from Paper G.

On that interpretation, the explanatory gap may be treated as structural rather than merely technical [7]. Establishing such a horizon would require a separate formal bridge from algebraic self-reference to phenomenal description; the analogy alone does not supply it.

This changes what counts as philosophical success. The goal is no longer to show how an independently specified physical world outputs experience. It is to understand why an embedded agent necessarily encounters a limit in its attempt to completely internalize the relation between structural invariants and lived articulation. The residual gap is relocated and reclassified, not removed.

8.3 Comparison with Existing Positions

The closest philosophical relatives of the present framework are positions that deny a brute separation between mind and world. Even so, the relation is exact only up to a point. The following comparison is necessarily compressed; each position listed has significant internal diversity, and the characterizations below refer to the canonical or most common formulation.

Position	Relation to the present paper
Physicalism / functionalism	Shares the desire for structural rigor, but typically preserves an explanatory asymmetry: physical facts are taken as independently specifiable, and consciousness must be shown to follow. The present theorem denies that asymmetry in its standard form.
Panpsychism	Avoids emergence by treating experience as basic, but inherits the combination problem. The present framework does not posit basic micro-experiences; it treats modular structure and geometry as co-emergent readings of one algebraic organization.
Dual-aspect monism	The nearest metaphysical relative. The decisive difference is that here the dual-aspect claim is not left at the level of metaphysical slogan; it is tied to a concrete inseparability theorem.
IIT	Also seeks a mathematically articulated bridge, but introduces an additional quantity Φ as a privileged measure. The present route instead treats inseparability as internal to modular and geometric structure, not as the effect of an extra consciousness functional.
Illusionism	Rejects the hard problem by denying that phenomenal consciousness needs the explanatory role usually assigned to it. The present framework does not deny experience. It denies the independent specification presupposition that makes experience look like a separately generated add-on.
Russellian monism	Very close in spirit insofar as it challenges the exhaustiveness of third-person structure. But Russellian monism often speaks of an intrinsic categorical basis behind structural physics. The present framework is more austere: both geometry and first-person time are readings of one algebraic structure, without appeal to a hidden intrinsic substrate beyond that structure.

The upshot is that the present work should not be advertised as a simple algebraic version of any single existing position. It is closest to a theorem-backed dual-aspect view, but even that label is only approximate. The real novelty lies in replacing a metaphysical balance claim with a rigidity statement.

A structural correspondence with the Yogacara fourfold analysis is recorded in Appendix A.

9 Broader Significance

The philosophical argument of the preceding sections is the primary contribution of this paper. The mathematical and physical content, however, has independent interest that is worth recording briefly.

9.1 Mathematical Significance

The Wiesbrock-GLW programme [17, 5] established that half-sided modular inclusions reconstruct conformal symmetry. What remained open was the passage from conformal skeleton to metric scale using only modular data. The present paper isolates a precise conditional route to a full geometric datum in 1+1 dimensions: the cross-ratio modular scale datum $\{K_b\}$ serves as the canonical non-gauge geometric modulus, and the Araki relative entropy Hessian [1] serves as the Type III-compatible QFIM that converts it into a Riemannian scale shadow. The route is conditional on $(LImp^1)$, $(CR\text{-}nd)$, and (SFa), all three of which are verified in two structurally distinct models. For Diff-covariant nets, Section 6 further shows that $(LImp^1)$ is automatic, leaving only $(CR\text{-}nd)$ and (SFa) as model-level verifications. In the worked examples, where the cross-ratio parameter naturally parametrizes the spatial slice, this datum assembles into the Lorentzian metric.

The universal scaling law $h_b \sim (1-b)^{-2}db^2$, shared by both worked examples, is a structural consequence of the Casini-Huerta-Myers form [3] of interval modular Hamiltonians. It is not a free-field accident. The model-dependent information enters only through the overall constant, which encodes data such as the central charge.

9.2 Physical Significance

The slogan "it from qubit" and the ER=EPR programme assert that entanglement structure determines spacetime geometry [11, 13, 16, 12]. These claims have so far been tied to the AdS/CFT correspondence. The present result establishes a version of the same thesis without assuming holographic duality. The axiom package selected by $(U3)$, together with the axiom package $(U1)+(C)+(Amb)+(CN_{\exists})+(LImp^1)+(CR\text{-}nd)+(SFa)$, makes no reference to anti-de Sitter space, conformal field theory on a boundary, or bulk-boundary correspondence. Inseparability is a property of Type III_1 algebraic structure under appropriate regularity [18], not a special consequence of holography.

This also resolves two open issues from the earlier HAFF programme (Paper D) [8]. The non-uniqueness of the map from algebraic deformations to state deformations is absorbed by the cross-ratio datum and the chart axiom, which together fix the relevant deformation direction. The signature mismatch between the positive-definite QFIM and the Lorentzian metric is resolved by restricting the QFIM to spatial slices and letting the conformal skeleton supply the causal structure.

10 Explicit Non-Claims and Open Problems

10.1 What This Paper Does Not Claim

1. It does not claim to have solved the hard problem of consciousness. It claims to have shown that the standard production formulation rests on a presupposition that is not available in the relevant mathematical framework.

2. It does not claim that modular flow "is" consciousness. The identification in Section 2 is a structural correspondence: modular flow plays the mathematical role that first-person temporal structure plays in the philosophical debate.

3. It does not claim that the inseparability theorem holds unconditionally for all Type III_1 factors. Phase 1 is unconditional within the stated axiom class. Phase 2 is conditional

on $(LImp^1)$, $(CR\text{-}nd)$, and (SFa). For Diff-covariant nets, $(LImp^1)$ is discharged by Theorem 6.1; the remaining two conditions are verified in two models but not proved in full generality.

4. It does not claim that the Buddhist correspondence in Section 7.4 participates in any proof. It is included as a structural comparison, not as evidence.

5. It does not claim that the results extend to dimensions higher than 1+1. The GLW construction produces a Möbius net on S^1. Higher-dimensional generalizations require additional HSMI directions and remain open.

6. It does not claim that consciousness is a quantum effect. The argument concerns the structural relationship between geometry and modular flow at the level of algebraic quantum theory, not a physical mechanism involving quantum coherence in biological systems.

10.2 Open Problems

1. Abstract sufficient conditions for $(CR\text{-}nd)$ and (SFa). Section 6 provides sufficient conditions for $(LImp^1)$ (Diff-covariance or split-assisted transportability), but the remaining two conditions are still verified only in specific models. A general criterion for $(CR\text{-}nd)$, perhaps in terms of spectral properties of the stress-energy tensor, would further reduce the conditionality.

2. Foliation compatibility beyond 1+1 dimensions. In the present 1+1 dimensional setting, the cross-ratio family is a single-parameter family and the assembly is straightforward. In higher dimensions, the scale data from different cross-ratio slices must satisfy nontrivial mutual consistency conditions. Formulating and verifying these conditions is an open problem tied to the higher-dimensional extension.

3. Higher-dimensional extension. In $d > 1 + 1$, the conformal group is larger and the cross-ratio construction must be replaced by a higher-dimensional analogue. Whether the inseparability mechanism survives this extension is the most important structural open question.

4. A formal self-description bridge. Paper G currently supplies philosophical motivation, not a fixed-point no-go theorem. A separate model with a typed syntax/observable interface and an explicit capacity functional would be required before modular invariants could yield a quantitative self-description bound.

5. The relationship between the present framework and Integrated Information Theory. Both seek mathematically articulated accounts of consciousness. Whether the inseparability theorem implies constraints on Φ or vice versa is unexplored.

A Yogacara Fourfold Correspondence

The Yogacara fourfold analysis provides a structural correspondence for the architecture that emerges in this paper. The correspondence does not enter any proof; it is included as a disciplined interpretive comparison that becomes newly intelligible once the mathematical theorem is in place. Each position listed below has significant internal diversity within the Buddhist philosophical tradition; the characterizations refer to the canonical formulation.

Layer	Phenomenological role	Mathematical counterpart
相分	the presented object-side	emergent geometry Geom
见分	the act-side of cognition	modular flow σ_t^{ω}
自证分	reflexive awareness	the self-calibrating informational loop of T-DOME Paper III [10]
证自证分	the boundary of reflexivity	the incompleteness limit of Paper G [9]

The structural fit is that object-side and act-side are not heterogeneous substances but internally linked aspects of a single cognitive event; reflexivity does not continue indefinitely into an infinite hierarchy, but closes through a higher-order boundary condition. This is exactly the pattern suggested by the inseparability framework: geometry and modular flow are inseparable at the base level, and reflexive closure is marked by incompleteness rather than by defect.

References

[1] H. Araki, *Relative entropy of states of von Neumann algebras*, Publ. Res. Inst. Math. Sci. Kyoto Univ. **11**, 809–833 (1976).

[2] H.-J. Borchers, *The CPT theorem in two-dimensional theories of local observables*, Commun. Math. Phys. **143**, 315–332 (1992).

[3] H. Casini, M. Huerta, and R. C. Myers, *Towards a derivation of holographic entanglement entropy*, JHEP **1105**, 036 (2011), arXiv:1102.0440.

[4] D. J. Chalmers, *The Conscious Mind: In Search of a Fundamental Theory*, Oxford University Press (1996).

[5] R. Guido, R. Longo, and H.-W. Wiesbrock, *Extensions of conformal nets and superselection structures*, Commun. Math. Phys. **192**, 217–244 (1998).

[6] R. Haag, *Local Quantum Physics: Fields, Particles, Algebras*, 2nd ed., Springer (1996).

[7] J. Levine, *Materialism and qualia: The explanatory gap*, Pacific Philosophical Quarterly **64**, 354 (1983).

[8] S. Liu, *Observable-Algebra Selection and Emergent Gravitational Phenomena: A Conditional Structural Analysis*, Zenodo (2026), DOI: 10.5281/zenodo.18388881.

[9] S. Liu, *Structural Limits of Unification: Accessibility, Incompleteness, and a Programme-Level Final-Cut Diagnosis*, Zenodo (2026), DOI: 10.5281/zenodo.18402907.

[10] S. Liu, *Fisher Information Geometry and Conditional Implementation Costs of Self-Referential Calibration*, Zenodo (2026), DOI: 10.5281/zenodo.18591771.

[11] J. M. Maldacena, *The large N limit of superconformal field theories and supergravity*, Int. J. Theor. Phys. **38**, 1113 (1999).

[12] J. Maldacena and L. Susskind, *Cool horizons for entangled black holes*, Fortsch. Phys. **61**, 781 (2013), arXiv:1306.0533.

[13] S. Ryu and T. Takayanagi, *Holographic derivation of entanglement entropy from AdS/CFT*, Phys. Rev. Lett. **96**, 181602 (2006).

[14] M. Takesaki, *Theory of Operator Algebras II*, Encyclopaedia of Mathematical Sciences **125**, Springer (2003).

[15] W. G. Unruh, *Notes on black-hole evaporation*, Phys. Rev. D **14**, 870 (1976).

[16] M. Van Raamsdonk, *Building up spacetime with quantum entanglement*, Gen. Rel. Grav. **42**, 2323 (2010).

[17] H.-W. Wiesbrock, *Half-sided modular inclusions of von Neumann algebras*, Commun. Math. Phys. **157**, 83 (1993).

[18] E. Witten, *APS Medal for Exceptional Achievement in Research: Invited article on entanglement properties of quantum field theory*, Rev. Mod. Phys. **90**, 045003 (2018), arXiv:1803.04993.

Higher-Dimensional Consistency of Geometry and Modular Flow: Small-Ball Modular Data and Linearized Gravity

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Abstract

Paper I established that, in 1+1 dimensions, emergent geometry and modular flow are not independently deformable. The higher-dimensional sequel cannot proceed by simply replacing the one-parameter cross-ratio family with a higher-dimensional analogue. For ball-shaped regions in $d+1$ dimensions, the relevant non-gauge datum is not an individual nested configuration but the full family of modular Hamiltonians attached to all balls and all local rest frames. The new question is therefore one of consistency: when can this multi-directional family of modular data be explained by a single geometry?

We study this question in the CHM ball framework for conformal nets with finite-dimensional conformal covariance, in the linearized regime. First, we show that the small-ball limit of the modular energy uniquely recovers the local stress tensor from the all-directions family of modular data. Second, for a local geometric functional satisfying linearity, locality, covariance, gauge invariance, and the correct scaling behavior, we additionally assume that its leading coefficient is a local symmetric rank-2 tensor of exactly two-derivative order and is divergence-free. Under those independently supplied premises, the linearized Lovelock classification identifies that coefficient with the linearized Einstein tensor up to normalization. Matching the modular and geometric small-ball expansions then yields the linearized Einstein equation within the stated domain. Thus, in higher dimensions, inseparability does not merely lock geometry and modular flow together; the consistency of that locking across directions takes the form of linearized gravitational constraint equations.

The resulting theorem is conditional but precise. It does not attempt full generality, nonlinear dynamics, or arbitrary Type III_1 nets. Its proof domain is restricted to CHM ball data in conformally covariant theories, in the linearized regime. Within that domain, however, the conclusion is sharp: the linearized Einstein equation appears as the consistency condition required for a single geometry to remain compatible with the full modular data family.

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1 Introduction

Paper I[1] showed that, in $1+1$ dimensions, the standard hard-problem presupposition of independent describability fails because geometry and modular flow cannot be independently deformed. That result depended on a one-parameter datum: the cross-ratio family of double intervals. In higher dimensions, the mathematical problem changes in a decisive way. There is no longer a single distinguished scalar modulus whose variation carries the entire scale information. The higher-dimensional sequel must therefore identify a different object.

The central claim of the present paper is that the correct object is the full family of ball modular Hamiltonians, indexed not only by center and radius but also by local timelike direction. Once this is recognized, the problem becomes a consistency problem rather than a parametrization problem. A single ball gives only one local modular datum. A family of balls in many directions gives many such data. The question is whether they can all be explained by one and the same geometry.

Our conditional answer is that, in the CHM ball framework and at linear order, consistency yields an Einstein coefficient equation once a local, two-derivative, divergence-free geometric coefficient is supplied. The CHM data alone recover the local stress tensor; they do not select the geometric operator.

The proof strategy is intentionally narrow. We do not attempt a nonlinear theorem, a fully general Type III_1 result, or an "if and only if" in complete abstractity. Instead we isolate a tractable domain in which the statement can be made cleanly: conformal nets with finite-dimensional conformal covariance, CHM modular Hamiltonians for balls, the vacuum or near-vacuum linearized regime, and a local geometric functional satisfying a small list of natural structural axioms. Within that domain, the theorem is short. The real work lies in identifying the correct small-ball expansions on both the modular and geometric sides.

The paper is organized in the reverse order of discovery. Section 2 defines the higher-dimensional modular datum and explains why the natural $1+1$-dimensional cross-ratio strategy has no direct higher-dimensional analogue. Section 3 formulates the small-ball consistency condition that replaces assembly in higher dimensions. Section 4 states and proves the main theorem. Section 5 records the free-field worked example that motivated the final theorem statements. Section 6 discusses the converse direction and the limits of the argument. Section 7 places the result in the broader Consciousness Trilogy, and Section 8 lists the natural open problems.

Remark 1.1 (Relation to Paper I). Paper II is mathematically self-contained. It uses Paper I only for motivation and context: the conceptual shift from "independent describability" to "independent deformability," and the observation that in $1+1$ dimensions the cross-ratio family carries the full scale information. No theorem from Paper I is invoked as a premise in any proof of Paper II. The CHM ball framework, the geometric axioms (G1)–(G5), and the small-ball consistency condition (MC) are all defined and used within the present paper.

2 Multi-Directional Modular Data in Higher Dimensions

2.1 From Cross-Ratio to Ball Families

Paper I[1] relied on the fact that, in $1+1$ dimensions, the correct non-gauge local modulus is the cross-ratio of a double-interval configuration. That datum is one-dimensional, and the resulting assembly problem is trivial. There is only one parameter direction, so there is no compatibility problem between different directional slices.

In higher dimensions, the naive analogue fails. A single ball is a conformal gauge orbit. Even a nested pair of balls does not produce a nontrivial scalar modulus in the way that a double interval does in $1+1$ dimensions. The correct higher-dimensional object therefore cannot be "a better cross-ratio." The no-go is structural: the relevant conformal degrees of freedom are not carried by one isolated nested configuration.

What replaces the cross-ratio is the family of all ball modular data. The local region is now a ball

$$B(\mathbf{x}_0, R) \subset \mathbb{R}^d,$$

or, more invariantly, a small causal diamond with center x_0, radius R, and rest-frame normal u^μ. Each such region gives one modular Hamiltonian. The higher-dimensional datum is not one scalar parameter but a family

$$\{K_{B,u}\}_{(B,u)\in\mathcal{B}},$$

where $\mathcal{B}$ denotes the set of all sufficiently small balls in all local rest frames.

2.2 CHM Ball Hamiltonians

In the vacuum of a conformal theory, the modular Hamiltonian of a ball is local and universal. In a rest frame adapted to the ball, the Casini–Huerta–Myers[2] formula gives

$$K_{B,u} = 2\pi \int_{B_u} d^d x \, \frac{R^2 - |\mathbf{x}-\mathbf{x}_0|^2}{2R} \, T_{\mu\nu}(x)\, u^\mu u^\nu.$$

The importance of this formula for the present paper is twofold. First, it supplies an explicit local kernel. Second, it shows that what the ball modular Hamiltonian probes is not an arbitrary operator but the rest-frame energy density $T_{uu} := T_{\mu\nu}u^\mu u^\nu$.

This already reveals the first higher-dimensional subtlety. If one restricts to a single fixed time slice, then the family of all balls only probes T_{00}. That is enough to recover the Hamiltonian constraint source, but not enough to recover the full stress tensor. To reconstruct the full tensorial source, the family must vary not only in center and radius but also in local timelike direction. This is why "multi-directional modular datum" is not rhetorical excess but a necessary mathematical strengthening.

2.3 The Canonical Datum

We therefore take the higher-dimensional canonical modular datum to be the full ball family

$$K_{B,u} := -\ln \Delta_{\mathcal{A}(B_u),\Omega},$$

understood in the CHM class where the modular Hamiltonian admits the local stress-tensor representation above. This datum is canonical in exactly the sense needed for the sequel to Paper I[1]: it is the minimal non-gauge family rich enough to encode the directional dependence that becomes nontrivial in dimensions higher than $1+1$.

The conceptual shift is worth making explicit. In Paper I[1], geometry and modular flow were locked by a one-parameter scale datum. In the present paper, they are locked by the requirement that an entire family of local modular energies be compatible with one geometry at once. What becomes nontrivial in higher dimensions is therefore not "finding the right modulus," but enforcing coherence across a family of moduli.

3 The Small-Ball Consistency Condition

3.1 Why Small Balls First

The strongest possible version of Paper II would formulate a global consistency condition on the modular data of all balls and derive the linearized Einstein equation from that global condition. We do not begin there. The small-ball regime already contains the local dynamical content, while avoiding the additional integrability problems of a global ball-space formulation. If the thesis fails even in the small-ball limit, the global version has no chance. If it succeeds there, the globalization problem becomes a genuine strengthening rather than a prerequisite.

3.2 Modular Small-Ball Expansion

Let $\delta\omega$ be a linearized perturbation of the vacuum, and assume the local stress tensor is smooth in a neighborhood of a point x_0. Then the CHM kernel gives an immediate small-ball asymptotic:

$$\delta\langle K_{B,u}\rangle = \frac{2\pi V_d}{d+2}\, R^{d+1}\, \delta\langle T_{\mu\nu}(x_0)\rangle u^\mu u^\nu + O(R^{d+3}),$$

where $V_d = \pi^{d/2}/\Gamma(d/2+1)$ is the volume of the unit d-ball. The key point is not the coefficient but the structural content: the leading small-ball term is the quadratic form of the local stress tensor evaluated on the timelike direction u^μ.

Because the family is taken over all timelike u^μ, the leading term uniquely determines the full symmetric tensor $\delta\langle T_{\mu\nu}(x_0)\rangle$. This is the first half of the main theorem. The all-directions small-ball modular datum is already strong enough to recover the complete local matter source.

3.3 Geometric Small-Ball Functional

The geometric side is intentionally abstract. We do not assume a particular holographic dictionary, RT formula[6], or bulk construction. Instead we assume that the geometry supplies, for each small ball and each direction, a linearized functional

$$\delta\mathcal{G}_{B,u}[h],$$

depending on the metric perturbation $h_{\mu\nu}$, subject to five structural requirements:

1. linearity in h;
2. locality in the small-ball limit;
3. covariance under local Lorentz transformations;
4. invariance under linearized gauge transformations;
5. the same leading scaling R^{d+1} as the modular side.

These assumptions are not arbitrary. They are exactly what is needed for a small-ball geometric quantity to be comparable to the CHM modular energy without smuggling in additional dynamics by hand.

3.4 Consistency

The higher-dimensional consistency condition can now be stated in its minimal form.

Small-ball modular consistency (MC). For all sufficiently small balls and all local timelike directions u^μ, the modular small-ball functional and the geometric small-ball functional agree:

$$\delta\mathcal{G}_{B,u}[h] = \delta\langle K_{B,u}\rangle.$$

This is the higher-dimensional replacement for the trivial assembly of Paper I[1]. In $1+1$ dimensions, there is only one parameter direction, so the scale datum needs no consistency check beyond existence. In higher dimensions, the requirement that all directional slices fit one geometry is itself the dynamical content.

4 Conditional Einstein Coefficient Matching

4.1 Modular Extraction Proposition

The modular side is summarized by the following statement.

Proposition 4.1. *In the CHM ball framework, for a linearized perturbation of the vacuum with a smooth local stress tensor, the all-directions small-ball modular datum has leading expansion*

$$\delta\langle K_{B,u}\rangle = \frac{2\pi V_d}{d+2}\, R^{d+1}\, \delta\langle T_{\mu\nu}(x_0)\rangle u^\mu u^\nu + O(R^{d+3}).$$

The proof is the small-ball Taylor expansion of the CHM kernel, together with the symmetry of the ball that kills the linear term. The resulting leading coefficient is universal and independent of the detailed model.

Corollary 4.2. *The all-directions leading small-ball term uniquely determines the full symmetric tensor $\delta\langle T_{\mu\nu}(x_0)\rangle$ at each point.*

This is immediate because a symmetric rank-2 tensor is uniquely fixed by its quadratic form on all timelike directions.

4.2 Missing Premises for Geometric Identification

The former claim that (G1)–(G5) alone select the linearized Einstein tensor is withdrawn. Those conditions do not restrict derivative order, and even at two derivatives the relevant coefficient must separately be required to be transverse. The following premises are therefore not consequences of (G1)–(G5):

(G-der2) The leading coefficient of $\delta\mathcal{G}_{B,u}[h]$ is a local symmetric rank-2 tensor, linear in h, of exactly two derivative order.

(G-div) That coefficient is divergence-free on the Minkowski background.

When (G-der2), (G-div), linearized gauge invariance, and Lorentz covariance are supplied, the standard linearized Lovelock classification identifies the coefficient with $\alpha_d G^{\mathrm{lin}}_{\mu\nu}$ up to normalization [5]. This is a conditional coefficient identification, not a uniqueness result derived from (G1)–(G5).

4.3 Conditional Coefficient Matching

With the missing geometric premises displayed, coefficient matching is immediate.

Theorem 4.3 (Conditional Einstein coefficient matching)**.** *Assume:*

1. *CHM modular Hamiltonians for all sufficiently small balls;*
2. *conformal covariance of the net (finite-dimensional conformal group* $\mathrm{SO}(d,2)$*);*
3. *linearization around a Minkowski vacuum background;*
4. *existence of a geometric functional satisfying the small-ball axioms (G1)–(G5) above;*
5. *the additional premises (G-der2) and (G-div);*
6. *small-ball modular consistency (MC).*

Then

$$G^{\mathrm{lin}}_{\mu\nu} = \kappa_d\, \delta\langle T_{\mu\nu}\rangle,$$

with $\kappa_d = \frac{2\pi V_d}{\alpha_d(d+2)}$.

The proof first uses the explicitly supplied (G-der2) and (G-div), together with gauge invariance and covariance, to invoke the linearized Lovelock classification. It then compares the resulting $G^{\mathrm{lin}}_{\mu\nu}u^\mu u^\nu$ coefficient with the CHM stress-tensor coefficient. Equality for all timelike u^μ forces equality of the tensors themselves.

Remark 4.4 (Conditionality and consistency). Assumptions (4)–(5) are the most significant: neither the geometric functional nor its two-derivative transverse leading coefficient is derived from the algebraic data. The theorem says that *if* such a functional exists and agrees with the modular side for all sufficiently small balls, *then* its leading local content is forced to be Einsteinian. The internal construction of $\delta\mathcal{G}_{B,u}$ from modular data alone remains the most significant open problem (Section 8.3).

The matching is also dynamically consistent: the linearized Einstein tensor satisfies the Bianchi identity $\partial^\mu G^{\mathrm{lin}}_{\mu\nu} = 0$, and the stress tensor satisfies the conservation law $\partial^\mu \delta\langle T_{\mu\nu}\rangle = 0$. Both sides of the equation are independently divergence-free.

4.4 Interpretation

This theorem is the higher-dimensional analogue of the inseparability theorem of Paper I[1], but with one crucial new feature. In $1+1$ dimensions, inseparability rules out independent deformation. In higher dimensions, the demand that the inseparable structures remain mutually consistent across all directions forces the geometry to obey linearized gravitational dynamics.

That is the sense in which gravity enters the trilogy. It is not introduced as an additional law attached from outside the modular-geometric relation. It appears as the consistency condition required for that relation to survive directional proliferation.

5 Worked Example: Free Conformal Scalar Field in $d+1$ Dimensions

This section verifies the main theorem in the simplest higher-dimensional model where all ingredients are explicit: the free massless conformal scalar field in $d+1$ dimensions. The verification has three parts: the stress tensor and its perturbation, the small-ball modular expansion, and the consistency check against the linearized Einstein equation.

5.1 The Model

Let ϕ be a free massless scalar field in $d+1$ Minkowski spacetime, with action

$$S = -\frac{1}{2}\int d^{d+1}x\,(\partial\phi)^2.$$

The stress-energy tensor is

$$T_{\mu\nu} = \partial_\mu\phi\,\partial_\nu\phi - \frac{1}{2}\eta_{\mu\nu}(\partial\phi)^2,$$

with the conformal improvement term understood when $d > 1$. In the vacuum state $|\Omega\rangle$, $\langle T_{\mu\nu}\rangle = 0$.

5.2 Perturbation: Classical Source

For the purpose of verifying the small-ball expansion, we do not need the full machinery of coherent states. It suffices to take any smooth, conserved classical stress tensor as the source. Let ϕ_{cl} be a smooth solution of the free wave equation $\Box\phi_{\text{cl}} = 0$, and define

$$\delta\langle T_{\mu\nu}(\mathbf{x})\rangle := \partial_\mu\phi_{\text{cl}}\,\partial_\nu\phi_{\text{cl}} - \frac{1}{2}\eta_{\mu\nu}(\partial\phi_{\text{cl}})^2,$$

with the conformal improvement understood when $d > 1$. This is the classical stress tensor of the on-shell configuration ϕ_{cl}. It is smooth, symmetric, and satisfies the conservation law

$$\partial^\mu\delta\langle T_{\mu\nu}\rangle = 0$$

as a direct consequence of $\Box\phi_{\text{cl}} = 0$.

The "linearized regime" of the theorem refers to linearization in the metric perturbation $h_{\mu\nu}$, not in the matter amplitude. The classical source $\delta\langle T_{\mu\nu}\rangle$ is a fixed, smooth tensor field that enters the right-hand side of the linearized Einstein equation. No conflict arises between the quadratic nature of the stress tensor in ϕ_{cl} and the linearized-gravity framework.

5.3 Modular Side: Small-Ball Expansion

For a ball $B(\mathbf{x}_0, R)$ in the rest frame with timelike normal $u^\mu = (1, \mathbf{0})$, the CHM formula gives

$$\delta\langle K_B\rangle = 2\pi\int_{|\mathbf{x}-\mathbf{x}_0|<R} d^dx\,\frac{R^2 - |\mathbf{x}-\mathbf{x}_0|^2}{2R}\,\delta\langle T_{00}(\mathbf{x})\rangle.$$

Expanding $\delta\langle T_{00}(\mathbf{x})\rangle$ around $\mathbf{x}_0$:

$$\delta\langle T_{00}(\mathbf{x})\rangle = \delta\langle T_{00}(\mathbf{x}_0)\rangle + \partial_i\delta\langle T_{00}\rangle(\mathbf{x}_0)(x^i - x_0^i) + O(|\mathbf{x}-\mathbf{x}_0|^2).$$

The linear term integrates to zero by the spherical symmetry of the ball. The leading contribution is therefore

$$\delta\langle K_B\rangle = 2\pi\,\delta\langle T_{00}(\mathbf{x}_0)\rangle \int_{|\mathbf{y}|<R} d^d y\,\frac{R^2-|\mathbf{y}|^2}{2R} + O(R^{d+3}).$$

The radial integral evaluates to

$$\int_{|\mathbf{y}|<R} d^d y\,\frac{R^2-|\mathbf{y}|^2}{2R} = \frac{\Omega_{d-1}}{2R}\int_0^R r^{d-1}(R^2-r^2)\,dr = \frac{\Omega_{d-1}}{2R}\left(\frac{R^{d+2}}{d}-\frac{R^{d+2}}{d+2}\right) = \frac{\Omega_{d-1}\,R^{d+1}}{d(d+2)},$$

where $\Omega_{d-1} = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ is the surface area of the unit $(d-1)$-sphere. Since $V_d = \frac{\Omega_{d-1}}{d}$, this simplifies to

$$\int_{|\mathbf{y}|<R} d^d y\,\frac{R^2-|\mathbf{y}|^2}{2R} = \frac{V_d\,R^{d+1}}{d+2}.$$

Therefore

$$\delta\langle K_B\rangle = \frac{2\pi V_d}{d+2}\,R^{d+1}\,\delta\langle T_{00}(\mathbf{x}_0)\rangle + O(R^{d+3}).$$

For a boosted ball with timelike normal u^μ, the same computation in the adapted frame gives

$$\delta\langle K_{B,u}\rangle = \frac{2\pi V_d}{d+2}\,R^{d+1}\,\delta\langle T_{\mu\nu}(\mathbf{x}_0)\rangle\,u^\mu u^\nu + O(R^{d+3}),$$

confirming Proposition 4.1 in the free-field model.

5.4 Full Tensor Recovery

In the free-field classical source, $\delta\langle T_{\mu\nu}\rangle$ is a smooth symmetric tensor field. By Corollary 4.2, the all-directions small-ball data uniquely determine this tensor at each point. In the present model, this can be verified directly: the quadratic form $u^\mu u^\nu\,\delta\langle T_{\mu\nu}(\mathbf{x}_0)\rangle$ for all timelike u determines $\delta\langle T_{\mu\nu}(\mathbf{x}_0)\rangle$ because the timelike cone spans the full symmetric tensor space.

5.5 Conservation Law

The classical stress tensor satisfies the conservation law

$$\partial^\mu \delta\langle T_{\mu\nu}\rangle = 0$$

as established in Section 5.2 from the on-shell condition $\Box\phi_{\text{cl}} = 0$. This is the necessary compatibility condition for the comparison with the linearized Einstein tensor, which satisfies the Bianchi identity $\partial^\mu G^{\text{lin}}_{\mu\nu} = 0$. In the free-field model, both sides of the would-be Einstein equation are independently divergence-free, so the matching is dynamically consistent and not merely algebraic.

5.6 Consistency Check

Suppose a linearized metric perturbation $h_{\mu\nu}$ is sourced by the free-field stress tensor via the semiclassical Einstein equation

$$G^{\text{lin}}_{\mu\nu}[h] = 8\pi G_N\,\delta\langle T_{\mu\nu}\rangle.$$

If one additionally supplies a geometric small-ball functional whose leading coefficient has the (G-der2) and (G-div) form used in Theorem 4.3, then it has leading term

$$\delta\mathcal{G}_{B,u}[h] = \alpha_d\, R^{d+1}\, G^{\mathrm{lin}}_{\mu\nu}(\mathbf{x}_0)\, u^\mu u^\nu + O(R^{d+3}) = \alpha_d \cdot 8\pi G_N\, R^{d+1}\, \delta\langle T_{\mu\nu}(\mathbf{x}_0)\rangle\, u^\mu u^\nu + O(R^{d+3}).$$

Comparing with the modular side:

$$\delta\langle K_{B,u}\rangle = \frac{2\pi V_d}{d+2}\, R^{d+1}\, \delta\langle T_{\mu\nu}(\mathbf{x}_0)\rangle\, u^\mu u^\nu + O(R^{d+3}).$$

The consistency condition (MC) requires these to be equal. Matching coefficients:

$$\alpha_d \cdot 8\pi G_N = \frac{2\pi V_d}{d+2},$$

which fixes the gravitational coupling in terms of the geometric normalization constant α_d and the spatial dimension d. This is not a derivation of G_N from first principles — the constant α_d is part of the geometric functional's normalization — but it shows that the coefficient comparison is compatible with the linearized Einstein equation: if such an equation holds, the modular and geometric small-ball data match at leading order.

5.7 What the Example Does and Does Not Show

The free-field example verifies three things:

1. The small-ball modular expansion (Proposition 4.1) is not a formal artifact. It is realized with explicit, computable coefficients in a standard QFT model.

2. The conservation law on the modular side matches the Bianchi identity on the geometric side, so the comparison is dynamically consistent.

3. The coefficient comparison is compatible with a semiclassical Einstein interpretation: if a metric perturbation satisfying the linearized Einstein equation exists, then the consistency condition (MC) is satisfied at leading order.

The example does not construct the geometric functional and therefore does not establish the existence premise of the conditional theorem. It assumes the semiclassical Einstein equation and checks compatibility of that assumption with the CHM modular coefficient. It does not show that (MC) is the unique route to the Einstein equation, nor does it address the converse direction.

6 Converse, Limits, and Relation to Faulkner–Jacobson

6.1 Why the Theorem Is Not an Equivalence

Theorem 4.3 is a conditional coefficient-matching theorem. It begins with the small-ball modular family and additionally assumes a single geometric explanation with the two-derivative and transverse premises (G-der2) and (G-div). Under those supplied premises it concludes that the leading coefficients obey the linearized Einstein equation. A reverse implication would require substantially more structure. One would need a faithful reconstruction map

$$h_{\mu\nu} \longmapsto \{\delta\mathcal{G}_{B,u}[h]\}_{(B,u)}$$

defined for all sufficiently small balls, with the correct normalization and covariance properties, and with agreement not only at leading order but at the level of the full ball-dependent functional. The linearized Einstein equation by itself does not provide such a map.

This distinction is not pedantic. The coefficient comparison in Section 4 identifies the leading local tensorial term,

$$G^{\mathrm{lin}}_{\mu\nu}(x_0) \propto \delta\langle T_{\mu\nu}(x_0)\rangle,$$

because equality for all timelike u^μ forces equality of the corresponding quadratic forms. But that argument is deliberately local and asymptotic. It does not control the $O(R^{d+3})$ remainder, it does not determine the full dependence of $\delta\mathcal{G}_{B,u}[h]$ on the ball parameters, and it does not show that every solution of the linearized Einstein equation arises from a geometric functional satisfying (G1)–(G5), (G-der2), and (G-div). The converse direction therefore belongs to a stronger future theorem, not to the present one.

6.2 Precise Relation to the Faulkner–Jacobson Line

The closest existing line of work runs from the CHM ball modular Hamiltonian through the entanglement first law to the Faulkner[3]–Jacobson[4] derivations of linearized gravity (see also [8]). The overlap is genuine and should be stated without exaggeration. In both settings, ball-shaped regions are essential, the CHM kernel supplies the universal local weighting, and the comparison across sufficiently small balls in all directions forces a rank-2 tensor equation whose leading local content is Einsteinian. At the level of small-ball tensor algebra, both routes exploit the same rigid fact: on a Minkowski background, the only local, symmetric, gauge-invariant, two-derivative tensor linear in $h_{\mu\nu}$ is the linearized Einstein tensor up to normalization [5].

The difference lies in what is taken as primitive. Faulkner-type arguments begin from an entropy variation on the geometric side, typically mediated by holographic ingredients such as the RT formula[6] and, in the quantum-corrected setting, JLMS[7]. The present paper begins instead from the family of modular Hamiltonians itself and asks under which supplied geometric conditions one geometry can explain that family. The geometric side is therefore left abstract: we do not assume a bulk dual, a surface-area formula, or an entanglement-entropy interpretation. We do assume a small-ball geometric functional satisfying (G1)–(G5), (G-der2), and (G-div). The result is thus explicit about where the geometric input enters.

It is therefore best to regard the two routes as overlapping but not identical. If one instantiates $\delta\mathcal{G}_{B,u}[h]$ by an entropy or area variation of the appropriate kind, then the present framework reproduces the local coefficient comparison of the Faulkner–Jacobson argument. Without such an instantiation, Theorem 4.3 remains a conditional consistency statement rather than an internal derivation of gravity.

6.3 Limits of the Present Result

The domain of validity should also be kept explicit. First, the modular side is restricted to the CHM ball class in conformally covariant theories, so the theorem does not apply to arbitrary Type III_1 nets. Second, the argument is local and small-ball. It extracts the leading tensorial content near each point, but it does not yet provide a global all-balls integrability theorem. Third, the analysis is linearized around a Minkowski vacuum background. Extension to other maximally symmetric backgrounds (de Sitter, anti-de Sitter) would require replacing $G^{\mathrm{lin}}_{\mu\nu}$ by the appropriate background-relative linearized Einstein operator; this is a natural but nontrivial generalization. Nothing in the present paper proves the full nonlinear Einstein equation.

Fourth, the geometric functional and the premises (G-der2) and (G-div) are axiomatized rather than internally derived from the algebraic data. This is the main remaining mathematical burden. The theorem shows that once these inputs exist and agree with the modular side for all sufficiently small balls, their leading coefficients match in Einstein form. It does not explain why an admissible algebraic model should supply those inputs. Globalization, nonlinear order, and internal construction are therefore genuine next problems rather than hidden assumptions.

7 Broader Physical and Conceptual Implications

Paper I[1] gives a conditional algebraic inseparability result. The present paper asks what an explicitly restricted higher-dimensional coefficient comparison adds to that programme.

In $1+1$ dimensions, the conditional inseparability statement is a rigidity result. In higher dimensions, if a geometric functional also satisfies (G-der2) and (G-div), all-direction consistency gives a linearized Einstein coefficient equation. The dynamics is therefore identified only inside this supplied geometric regime.

This has three consequences for the trilogy as a whole.

First, it isolates the extra premises required to obtain an Einstein coefficient. The equation is not derived from modular consistency alone: the two-derivative and divergence-free geometric structure is supplied. Within that regime, all-direction consistency fixes the coefficient matching.

Second, it gives a conditional higher-dimensional extension of Paper I's programme. The gravitational coefficient equation follows only within the stated CHM, linearized, and supplied-geometric regime; it is not a general dynamical consequence of modular data.

Third, the $O(R^{d+3})$ remainder remains uncontrolled. This paper does not construct coefficient directions, a Fisher map, or a self-description budget from that remainder. Accordingly it supplies no remainder-to-self-description bridge for Paper III.

8 Open Problems

8.1 Globalization

The present theorem is small-ball and local. It extracts the leading tensorial content near each point but does not provide a global all-balls integrability theorem. A global version would require controlling the full ball-dependent functional $\delta\mathcal{G}_{B,u}[h]$, not just its small-ball asymptotic. This is a genuine strengthening that would bring the result closer to the Faulkner[3]–Jacobson[4] line, where the entanglement first law is applied to all balls simultaneously.

8.2 Nonlinear Order

The $O(R^{d+3})$ and higher terms in the small-ball expansion may encode nonlinear corrections to the Einstein equation. In the Jacobson[4] thermodynamic derivation, the passage from linearized to full nonlinear Einstein equations requires an additional equilibrium or maximal-entropy assumption. Whether a similar upgrade is available in the present modular-consistency framework is an open question. This is the natural bridge to a later gravity paper, not something to force into the present one.

8.3 Internal Construction of the Geometric Functional

At present, the geometric small-ball functional $\delta\mathcal{G}_{B,u}[h]$, (G-der2), and (G-div) are axiomatized, not constructed from purely algebraic data. The theorem identifies an Einstein coefficient only after those inputs are supplied. Constructing the functional and deriving its derivative order and transversality from modular data would be required to remove this conditionality.

8.4 No Current Coefficient-Tower Bridge

The higher-order remainder is only an asymptotic error term in the present analysis. No independent coefficient directions or Fisher map are constructed here, so it cannot be passed to Paper III as a physical control tower or quantitative self-description deficit. Establishing such a bridge would require new definitions and estimates beyond the current small-ball result.

8.5 Relation to Entanglement Equilibrium

Jacobson's[4] entanglement equilibrium approach derives the full nonlinear Einstein equation from the assumption that the vacuum maximizes entanglement entropy for small causal diamonds. The present framework does not assume entropy maximization. Whether the modular consistency condition (MC) implies, or is implied by, an entanglement equilibrium condition is an unexplored connection that could unify the two approaches.

References

[1] S. Liu, *The Inseparability of Geometry and Modular Flow: A Structural Inseparability Theorem and Its Implications for the Hard Problem*, Zenodo (2026), DOI: 10.5281/zenodo.19607387.

[2] H. Casini, M. Huerta, and R. C. Myers, *Towards a derivation of holographic entanglement entropy*, JHEP **1105**, 036 (2011), arXiv:1102.0440.

[3] T. Faulkner, M. Guica, T. Hartman, R. C. Myers, and M. Van Raamsdonk, *Gravitation from entanglement in holographic CFTs*, JHEP **1403**, 051 (2014), arXiv:1312.7856.

[4] T. Jacobson, *Entanglement equilibrium and the Einstein equation*, Phys. Rev. Lett. **116**, 201101 (2016), arXiv:1505.04753.

[5] D. Lovelock, *The Einstein tensor and its generalizations*, J. Math. Phys. **12**, 498–501 (1971).

[6] S. Ryu and T. Takayanagi, *Holographic derivation of entanglement entropy from AdS/CFT*, Phys. Rev. Lett. **96**, 181602 (2006).

[7] D. L. Jafferis, A. Lewkowycz, J. Maldacena, and S. J. Suh, *Relative entropy equals bulk relative entropy*, JHEP **1606**, 004 (2016), arXiv:1512.06431.

[8] N. Lashkari, M. B. McDermott, and M. Van Raamsdonk, *Gravitational dynamics from entanglement "thermodynamics"*, JHEP **1404**, 195 (2014), arXiv:1308.3716.

Finite Self-Models and the Open Explanatory Gap: A Conditional Arithmetic Analysis

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Abstract

Paper G offers a programme-level diagnosis of possible self-reference limits, while Paper II retains a leading small-ball stress-tensor extraction with an $O(R^{d+3})$ remainder. Neither result supplies a fixed-point no-go theorem, a physical Fisher spectrum, or a self-description tower. This paper tests a proposed quantitative bridge by defining a local Araki-relative-entropy reconstruction loss for a stipulated finite statistical bottleneck.

The former local minimax drift-floor theorem is withdrawn. A finite Fisher ceiling does not by itself imply a positive reconstruction loss: on a short modular orbit, an admissible two-outcome channel can preserve the orbit coordinate exactly and achieve zero loss. A future positive risk theorem would therefore need a typed observation model that excludes such lossless-orbit constructions and relates estimator risk to the reconstruction loss.

The surviving formal result is a conditional arithmetic inversion. If a stipulated positive sequence satisfies

$$\log I_k = -\alpha k \log k + O(k)$$

and the available budget grows at polynomial scale, then its maximal admissible index obeys

$$K(n) = O\left(\frac{\log n}{\log\log n}\right).$$

This lemma proves neither a physical coefficient/Fisher spectrum nor a self-model resource bound. The explanatory gap therefore remains open: the paper identifies conditions required for a future quantitative rebuild but does not establish a finite self-description horizon.

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1 Introduction

Paper G[3] motivates the possibility that self-reference limits an accessibility-based description. Its regress and collapse scenarios are philosophical diagnostics, not a fixed-point no-go theorem or an exhaustive dichotomy. Paper III uses that discussion only as motivation.

Paper II[2] moved the discussion to a more concrete level. In the higher-dimensional CHM ball framework, Paper II retains the leading stress-tensor extraction at order R^{d+1}, with a remainder $O(R^{d+3})$. It does not construct independent coefficient directions, a Fisher map, or a self-description tower from that remainder.

The present paper defines a local Araki-relative-entropy reconstruction loss for a stipulated finite statistical bottleneck, but withdraws the claimed universal drift floor. Estimation risk in a specified noisy experiment cannot be identified with reconstruction loss for a channel that may preserve the orbit coordinate exactly. The surviving formal result is instead a conditional arithmetic inversion for a stipulated positive sequence and budget. The surrounding programme draws on:

1. the BKM/QFIM structure already used in Paper I[1];
2. the self-referential Fisher and ego-rigidity formalism of T-DOME Paper III[4];
3. the leading CHM expansion and remainder order recorded in Paper II[2].

The arithmetic statement assumes, rather than derives, a positive sequence I_k with

$$\log I_k = -\alpha k \log k + O(k),$$

and a polynomial-scale budget. Under those stipulations its maximal index obeys

$$K(n) = O\left(\frac{\log n}{\log\log n}\right).$$

This proves neither a physical Fisher spectrum nor a self-model budget.

The trilogy-level interpretation also depends on Paper I's supplied shell interface (Amb), its dynamical covariance axiom (C), and the scale-faithfulness axiom (SFa). Modular flow supplies algebraic time only; a phenomenal-time reading is conditional and interpretive. T-DOME Paper III supplies an ego-rigidity prior model, not a theorem that turns it into a reconstruction floor.

The paper is organized as follows. Section 2 defines finite observers, admissible self-models, and the self-description deficit. Section 3 records why the former drift-floor theorem is withdrawn. Section 4 proves the conditional arithmetic inversion. Section 5 retains only the exact CHM moment coefficient and explains why it does not supply a Fisher spectrum. Section 6 explains the trilogy-level significance of the result, and Section 7 states the remaining limitations and open problems.

2 Finite Observers and the Self-Description Deficit

2.1 Finite Observers

Motivated by Paper G's architectural discussion, we first specify what counts as a finite statistical self-model. This definition does not inherit a no-go theorem from Paper G.

Definition 2.1 (Finite observer)**.** A finite observer of complexity n is a triple

$$(\mathcal{A}_{\text{obs}}, \omega_{\text{obs}}, \mathfrak{M}_n),$$

where:

1. $\mathcal{A}_{\text{obs}} \subset \mathcal{A}_c$ is the observer's accessible subalgebra;
2. $\omega_{\text{obs}} = \omega|_{\mathcal{A}_{\text{obs}}}$ is the restricted state;
3. $\mathfrak{M}_n$ is an n-dimensional statistical model used internally to represent the observer's own accessible state.

The complexity parameter n measures effective statistical degrees of freedom, not arbitrary description length. This is deliberate. Paper III is not about Kolmogorov complexity in full generality. It is about the finite number of statistically usable directions available to an observer trying to model itself.

2.2 Modular Fisher Sensitivity

The Tomita–Takesaki modular automorphism group σ_t^ω of a faithful normal state ω on a von Neumann algebra satisfies $\omega \circ \sigma_t^\omega = \omega$ for all t. The orbit is therefore trivial at the level of states, and the associated tangent vector vanishes. Since the modular orbit itself carries no state-level Fisher information, we replace it with the intrinsic accessible-unitary perturbation family as the operational proxy for modular sensitivity.

Definition 2.2 (Modular Fisher sensitivity)**.** Fix a self-adjoint element $h \in \mathcal{A}_{\text{obs}}$ and define the perturbed family

$$\omega_s^{(h)} := \omega_{\text{obs}} \circ \text{Ad}(e^{ish}), \qquad s \in (-\epsilon, \epsilon).$$

The modular Fisher sensitivity of ω_{obs} along h is

$$\mathcal{I}_F(\omega_{\text{obs}}, h) := \mathbf{Q}_{\omega_{\text{obs}}}\left(\left.\frac{d\omega_s^{(h)}}{ds}\right|_{s=0}, \left.\frac{d\omega_s^{(h)}}{ds}\right|_{s=0} \right).$$

The modular Fisher sensitivity of the observer is

$$\mathcal{I}_F(\omega_{\text{obs}}) := \sup\big\{ \mathcal{I}_F(\omega_{\text{obs}}, h) : \ h = h^* \in \mathcal{A}_{\text{obs}}, \ \|h\| \le 1 \big\}.$$

The supremum is taken over unit-norm self-adjoint perturbations in the observer's own accessible algebra. It is finite whenever the BKM form is bounded on the unit ball of $\mathcal{A}_{\text{obs}}$-tangent directions, which holds in the local regime used throughout this paper. The quantity $\mathcal{I}_F(\omega_{\text{obs}})$ measures the observer's maximal intrinsic sensitivity to unitary perturbations of its own accessible state. It is strictly positive whenever ω_{obs} is not a trace state on $\mathcal{A}_{\text{obs}}$.

Throughout the paper, the term *local modular drift family* refers to the one-parameter family $\{\omega_s^{(h)}\}$ for a fixed perturbation h. When the supremum in Definition 2.2 is attained, we take h to be a maximizer; when it is not, we use a standard η-optimization: for each $\eta > 0$, choose $h_\eta = h_\eta^* \in \mathcal{A}_{\text{obs}}$ with $\|h_\eta\| \le 1$ and $\mathcal{I}_F(\omega_{\text{obs}}, h_\eta) \ge \mathcal{I}_F(\omega_{\text{obs}}) - \eta$, carry out the proof for h_η, and let $\eta \downarrow 0$. Despite the name, this family is not the Tomita–Takesaki modular orbit (which is state-preserving) but the accessible-unitary perturbation family that serves as its operational proxy at the level of Fisher information. This is the family along which the observer's self-description is hardest to compress.

2.3 Admissible Self-Models

The theorem chain does not require a literal embedding of the self-model into a matrix algebra. That stronger route would force us to solve a much harder operator-algebraic problem than the present paper needs. What matters here is the existence of a finite statistical bottleneck.

Definition 2.3 (Admissible self-model)**.** An admissible n-dimensional self-model is a pair

$$(\Pi_n, \Psi_n),$$

where:

1. $\Pi_n : \mathcal{S}(\mathcal{A}_{\text{obs}}) \to \mathfrak{M}_n$ is the state-space map induced by a normal unital completely positive map from a finite-dimensional algebra into $\mathcal{A}_{\text{obs}}$, so that Petz monotonicity of relative entropy applies;
2. $\Psi_n : \mathfrak{M}_n \to \mathcal{S}(\mathcal{A}_{\text{obs}})$ is a recovery map;
3. Π_n factors through the observer's own accessible state space, so no external "God's-eye" parameter is introduced;
4. near the reference state ω_{obs}, the Fisher matrix on $\mathfrak{M}_n$ can be whitened:
$$\mathbf{I}_\omega^{(n)} = \text{diag}(\lambda_1, \ldots, \lambda_n), \qquad \lambda_a \le \mathcal{I}_F(\omega_{\text{obs}}).$$

The last condition is the substantive standing assumption of the theorem, not a mere coordinate normalization. It says that the compressed model has only n effective Fisher directions, and none of them is sharper than the observer's own modular sensitivity scale. This is the exact place where the theorem absorbs finite-dimensionality.

2.4 Self-Description Deficit

The correct loss function is not a metric but a divergence. In a Type III setting there is no preferred trace-class density-matrix picture, while Araki relative entropy is native to the operator algebraic formulation [5].

Definition 2.4 (Self-description deficit)**.** Define the BKM ball of radius δ around ω_{obs} by

$$\mathcal{B}_\delta(\omega_{\text{obs}}) := \big\{\omega \in \mathcal{S}(\mathcal{A}_{\text{obs}}) : \ \mathbf{Q}_{\omega_{\text{obs}}}(\omega - \omega_{\text{obs}},\, \omega - \omega_{\text{obs}}) < \delta^2\big\}.$$

Fix once and for all a $\delta > 0$ small enough that Araki relative entropy admits its quadratic expansion on $\mathcal{B}_\delta(\omega_{\text{obs}})$, and large enough that the ego-rigidity prior π_{ego} of T-DOME Paper III[4] can be supported on a segment of the local modular drift family inside $\mathcal{B}_\delta(\omega_{\text{obs}})$. The self-description deficit of complexity n is

$$\varepsilon(n) := \inf_{(\Pi_n, \Psi_n)} \sup_{\omega \in \mathcal{B}_\delta(\omega_{\text{obs}})} S(\omega \parallel \Psi_n \circ \Pi_n(\omega))\,,$$

where the infimum is taken over all admissible n-dimensional self-models.

Near a fixed state, Araki relative entropy has the familiar BKM/QFIM quadratic expansion

$$S(\omega \| \omega + \delta\omega) = \frac{1}{2}\mathbf{Q}_\omega(\delta\omega, \delta\omega) + o(\|\delta\omega\|^2),$$

which is the local information-geometric structure already used in Paper I[1] and in Petz's monotone-metric framework [7]. The minimax form is essential: it rules out trivial constant recovery maps that memorize the reference state while failing arbitrarily nearby. Thus $\varepsilon(n)$ measures optimal local uniform reconstruction loss in the intrinsic Type III geometry of states.

3 The Fisher Ceiling for Admissible Self-Models

3.1 Petz Monotonicity and Finite Fisher Budget

The first ingredient is standard.

Lemma 3.1 (Petz monotonicity)**.** *For any normal coarse-graining* Π_n*,*

$$S(\omega\|\sigma) \geq S(\Pi_n(\omega)\|\Pi_n(\sigma)).$$

Proof. This is the monotonicity of relative entropy under coarse-graining [6]. The present paper uses it only in the local regime, where it implies that compression cannot increase BKM/Fisher information. □

Whitening then converts finite statistical dimension directly into two distinct ceilings: one for the specific modular drift channel, and one for the total trace budget across all retained directions.

Proposition 3.2 (Directional and total Fisher ceilings)**.** *Let* (Π_n, Ψ_n) *be an admissible* n*-dimensional self-model. Then:*

1. *along the local modular drift family through* ω_{obs}*, the Fisher information retained after compression satisfies*
$$\mathcal{I}^{(n)}_{\text{drift}} \leq \mathcal{I}_F(\omega_{\text{obs}});$$
2. *the total retained data Fisher information satisfies*
$$\mathcal{I}^{(n)}_{\text{data}} := \operatorname{tr} \mathbf{I}^{(n)}_\omega \leq n\, \mathcal{I}_F(\omega_{\text{obs}}).$$

Proof. (1) Apply Lemma 3.1 to the one-parameter modular drift family. In the local quadratic regime, monotonicity of relative entropy implies monotonicity of the associated BKM/Fisher metric, so compression cannot increase Fisher information for that specific scalar parameter.

(2) In Fisher-whitened coordinates, the compressed Fisher matrix is

$$\mathbf{I}^{(n)}_\omega = \operatorname{diag}(\lambda_1, \ldots, \lambda_n), \qquad \lambda_a \leq \mathcal{I}_F(\omega_{\text{obs}}).$$

Summing over the n effective directions gives

$$\operatorname{tr} \mathbf{I}^{(n)}_\omega = \sum_{a=1}^{n} \lambda_a \leq n\, \mathcal{I}_F(\omega_{\text{obs}}).$$

□

3.2 Withdrawal of the Drift-Floor Claim

The former local minimax drift-floor theorem and its corollary are withdrawn. The argument applied a van Trees estimation-risk bound to a compression/recovery loss without specifying a noisy observation model that prevents the orbit coordinate from being retained.

The obstruction can be implemented by an internal finite-dimensional channel. Let $s \mapsto \omega_s$ be a nonconstant smooth local drift orbit. Choose an effect $0 \leq M \leq \mathbf{1}$ in $\mathcal{A}_{\text{obs}}$ with $\frac{d}{ds}\omega_s(M)|_{s=0} \neq 0$. The normal unital completely positive map

$$\Phi : \mathbb{C}^2 \longrightarrow \mathcal{A}_{\text{obs}}, \qquad \Phi(a, b) = aM + b(\mathbf{1} - M),$$

induces the binary statistical coordinate

$$\Pi_1(\omega_s) = \big(p(s), 1 - p(s)\big), \qquad p(s) := \omega_s(M).$$

By the inverse-function theorem, after restricting to a sufficiently short orbit segment, p is injective. Define the recovery map on its image, and extend it arbitrarily off that image, by

$$\Psi_1\big(p(s), 1 - p(s)\big) = \omega_s.$$

This code uses an observable in the accessible algebra rather than an external parameter, and it factors through a finite-dimensional commutative algebra as required by Definition 2.3. Then

$$S(\omega_s \| \Psi_1 \circ \Pi_1(\omega_s)) = 0$$

for every s on the encoded segment. A positive Bayes estimation risk can arise only after a likelihood or sampling channel restricts access to s; it is not a lower bound on this lossless orbit reconstruction. The previous proof therefore confused estimation risk with reconstruction loss. Petz monotonicity and the stipulated Fisher ceilings above remain bookkeeping facts within their stated model, but they imply no nonzero reconstruction floor.

4 A Conditional Arithmetic Inversion

Paper II[2] supplies a leading CHM coefficient and an $O(R^{d+3})$ remainder, not an infinite family of normalized Fisher directions. We therefore separate the arithmetic statement from any physical interpretation.

Let $\{I_k\}_{k\geq 1}$ be a *stipulated* positive sequence and let $B(n) \geq 1$ be a *stipulated* budget. Define

$$K(n) := \max\left\{ K \geq 1 : \sum_{j=1}^{K} I_j^{-1} \leq B(n) \right\},$$

with $K(n) = 0$ if the set is empty. No claim is made here that the I_k are Fisher informations of physical coefficient directions or that $B(n)$ is a self-model resource.

Lemma 4.1 (Conditional arithmetic inversion)**.** *Suppose*

$$\log I_k = -\alpha k \log k + O(k), \qquad \alpha > 0,$$

and $\log B(n) = O(\log n)$. *Then*

$$K(n) = O\left(\frac{\log n}{\log\log n}\right).$$

Proof. If $K = K(n) \geq 1$, the final nonnegative summand satisfies $I_K^{-1} \leq B(n)$. Hence

$$\alpha K \log K + O(K) = -\log I_K \leq \log B(n) = O(\log n).$$

For sufficiently large K, the $K \log K$ term dominates the linear remainder, so inversion gives $K = O(\log n / \log\log n)$. Bounded K obeys the same asymptotic bound. □

This lemma is exactly the surviving claim. It establishes neither a self-model budget nor a physical Fisher spectrum. In particular, rescaling coefficient coordinates changes any candidate I_k, so a physical application requires a separately justified normalization, likelihood, and Fisher map.

5 Exact CHM Moment Without a Fisher-Spectrum Claim

The following exact integral is retained as an algebraic check. It does not construct the coefficient directions or Fisher map absent from Paper II.

5.1 Exact d=1 Signal Factor

In $d = 1$, the CHM modular Hamiltonian is

$$K_B = \frac{\pi}{R} \int_{-R}^{R} (R^2 - x^2)\, T_{00}(x)\, dx.$$

Expanding $T_{00}(x)$ around the ball center gives

$$T_{00}(x) = \sum_{k \geq 0} \frac{x^{2k}}{(2k)!} \partial_x^{2k} T_{00}(0) + \cdots .$$

Therefore the coefficient multiplying the $2k$-th derivative is

$$c_k^{(d=1)} = \frac{\pi}{R} \cdot \frac{1}{(2k)!} \int_{-R}^{R} (R^2 - x^2) x^{2k}\, dx \cdot R^{-(2k+2)}.$$

The radial moment is exact:

$$\int_{-R}^{R} (R^2 - x^2) x^{2k}\, dx = \frac{4R^{2k+3}}{(2k+1)(2k+3)}.$$

Hence

$$c_k^{(d=1)} = \frac{4\pi}{(2k+1)(2k+3)(2k)!},$$

which shows exact factorial suppression:

$$c_k^{(d=1)} \sim \frac{\text{const}}{(2k)!\, k^2}.$$

The factorial suppression of $c_k^{(d=1)}$ is normalization-dependent signal information. It does not by itself determine a fluctuation kernel, a renormalized Fisher information I_k, or a cumulative inverse-information budget. Consequently this calculation does not instantiate Lemma 4.1; it only shows what one factor in a future, fully specified model would look like.

6 Trilogy Implications

6.1 Paper G as Motivation Only

Paper G[3] supplies philosophical motivation for studying self-description, but no formal bridge maps its syntax-level scenarios to the reconstruction loss defined here. The withdrawn drift-floor argument does not quantify Paper G, and this paper asserts no generic fixed-point obstruction.

6.2 Relocating the Gap in Paper II Terms

Paper II[2] supplies a leading CHM stress-tensor term and an $O(R^{d+3})$ remainder. It does not supply normalized coefficient directions or a Fisher map. Paper III therefore does not reinterpret the remainder as a physical control tower. Lemma 4.1 applies only after a positive sequence and budget are stipulated independently.

6.3 Trilogy-Level Interpretation

The trilogy-level reading is conditional on Paper I's shell interface (Amb), dynamical axiom (C), and scale-faithfulness assumption (SFa), as well as Paper II's restricted CHM regime. These inputs supply algebraic modular time and conditional geometric statements, not a phenomenal-time identification. The surviving arithmetic lemma does not locate or quantify an explanatory gap.

7 Limitations and Open Problems

The withdrawals identify four concrete tasks for any future rebuild.

1. **A typed observation model.** A future risk theorem must specify a likelihood or sampling channel that prevents lossless orbit coding, then distinguish estimator risk from state reconstruction loss.

2. **Normalized coefficient directions.** A physical use of Lemma 4.1 requires independently constructed coefficient perturbations, a renormalized kernel, and a normalization under which the Fisher sequence is invariantly defined.

3. **A Paper II bridge.** The remainder order alone is insufficient. One must construct a map from small-ball coefficient data to the statistical model before any resource comparison is meaningful.

4. **Ego-rigidity interface.** The T-DOME prior may enter a specified Bayesian estimation problem, but it supplies no reconstruction floor by itself. Its operational channel and boundary conditions must be given before van Trees can be used.

8 Conclusion

Paper III tests a proposed quantitative route to the explanatory gap. Paper G supplies programme-level motivation, while Paper II supplies a leading stress-tensor extraction and a controlled remainder order; it does not identify that remainder as a physical self-description carrier. The present paper defines a finite-self-model reconstruction loss and records why the former universal drift-floor inference fails.

A finite Fisher ceiling alone does not imply a nonzero local minimax reconstruction loss. The lossless-orbit construction shows that a finite channel can preserve the relevant modular-orbit coordinate exactly on a short segment. A positive lower bound may still arise under an additional typed observation model that excludes such encodings and connects estimation risk to reconstruction loss, but a precise characterization of those conditions remains open.

Separately, the conditional arithmetic lemma shows that a stipulated positive sequence with $\log I_k = -\alpha k \log k + O(k)$ has maximal admissible index $K(n) = O(\log n / \log\log n)$ under a

polynomial-scale budget. It does not establish a physical Fisher spectrum, a self-model budget, or a finite self-description horizon. The explanatory gap is therefore not eliminated in general, but it is not quantitatively located by the present results.

A Trilogy Key Premises Summary

The following list records the external inputs and their restricted roles. Paper III is not self-contained with respect to these trilogy interfaces.

1. **Paper I algebraic interface** (Paper I[1]). The Araki relative entropy of two nearby faithful normal states on a Type III von Neumann algebra admits a local quadratic expansion $S(\omega\|\omega+\delta\omega) = \frac{1}{2}\mathbf{Q}_\omega(\delta\omega,\delta\omega) + o(\|\delta\omega\|^2)$, where $\mathbf{Q}_\omega$ is the BKM/QFIM quadratic form. This is the information-geometric structure used in Definitions 2.2 and 2.4 and throughout Section 3. The broader trilogy interpretation also assumes Paper I's shell interface (Amb), dynamical covariance (C), and scale-faithfulness (SFa). These give conditional algebraic structure; modular flow is algebraic time, not a derived phenomenal-time variable.

2. **Small-ball expansion with $O(R^{d+3})$ remainder** (Paper II[2]). In the CHM ball framework for conformal nets, the modular energy of a small ball of radius R admits the expansion $\delta\langle K_{B,u}\rangle = \frac{2\pi V_d}{d+2} R^{d+1}\delta\langle T_{\mu\nu}\rangle u^\mu u^\nu + O(R^{d+3})$. The leading coefficient extracts the local stress tensor, while the higher-order remainder is uncontrolled. Paper II supplies neither an infinite coefficient tower nor a Fisher map; Section 5 uses only a separate exact $d = 1$ moment calculation.

3. **Self-reference motivation** (Paper G[3]). Paper G discusses regress and collapse scenarios but does not prove that a self-referential fixed point generically fails to exist. Paper III uses it as motivation only and does not quantify it.

4. **Ego-rigidity prior** (T-DOME Paper III[4]). The ego-rigidity prior $\mathcal{I}_{\text{ego}}$ is the Fisher information of the observer's prior distribution over its own self-model parameters: $\mathcal{I}_{\text{ego}} := \int(\partial_\sigma \log \pi_{\text{ego}}(\sigma))^2 \pi_{\text{ego}}(\sigma)\, d\sigma$. It may enter a separately specified Bayesian likelihood model. It does not lower-bound reconstruction loss without such an observation model, and no theorem in this paper uses it to do so.

References

[1] S. Liu, *The Inseparability of Geometry and Modular Flow: A Structural Inseparability Theorem and Its Implications for the Hard Problem*, Zenodo (2026), DOI: 10.5281/zenodo.19607387.

[2] S. Liu, *Higher-Dimensional Consistency of Geometry and Modular Flow: Small-Ball Modular Data and Linearized Gravity*, Zenodo (2026), DOI: 10.5281/zenodo.19607543.

[3] S. Liu, *Structural Limits of Unification: Accessibility, Incompleteness, and a Programme-Level Final-Cut Diagnosis*, Zenodo (2026), DOI: 10.5281/zenodo.18402907.

[4] S. Liu, *Fisher Information Geometry and Conditional Implementation Costs of Self-Referential Calibration*, Zenodo (2026), DOI: 10.5281/zenodo.18591771.

[5] H. Araki, *Relative entropy of states of von Neumann algebras*, Publ. Res. Inst. Math. Sci. **11**, 809–833 (1976).

[6] D. Petz, *Sufficient subalgebras and the relative entropy of states of a von Neumann algebra*, Commun. Math. Phys. **105**, 123–131 (1986).

[7] D. Petz, *Monotone metrics on matrix spaces*, Linear Algebra Appl. **244**, 81–96 (1996).

[8] H. L. Van Trees, *Detection, Estimation, and Modulation Theory, Part I*, Wiley, New York (1968).

[9] H. Casini, M. Huerta, and R. C. Myers, *Towards a derivation of holographic entanglement entropy*, JHEP **1105**, 036 (2011), arXiv:1102.0440.